

# Criptografía: Conceptos Fundamentales

Sebastián Giraudo

Por qué necesitamos sistemas de prueba?

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  - Estados intermedios de un cómputo



# Por qué necesitamos sistemas de prueba?

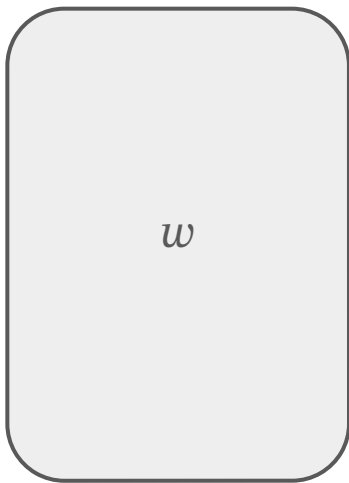
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Un **prover** quiere convencer a un **verifier** de que cierta afirmación es cierta.

La información que hace cierta una afirmación es el **witness**.

# Primer intento: enviar el witness

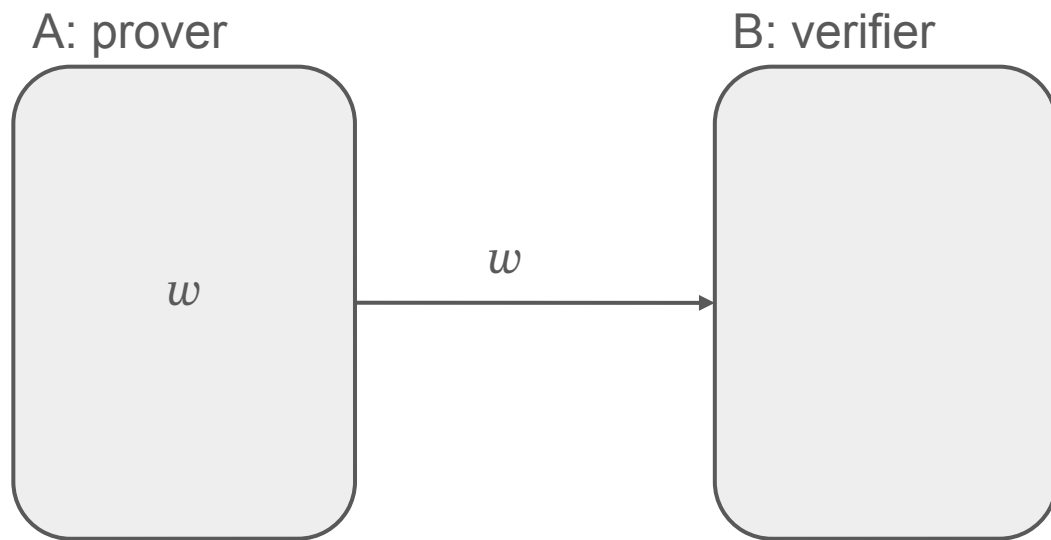
A: prover



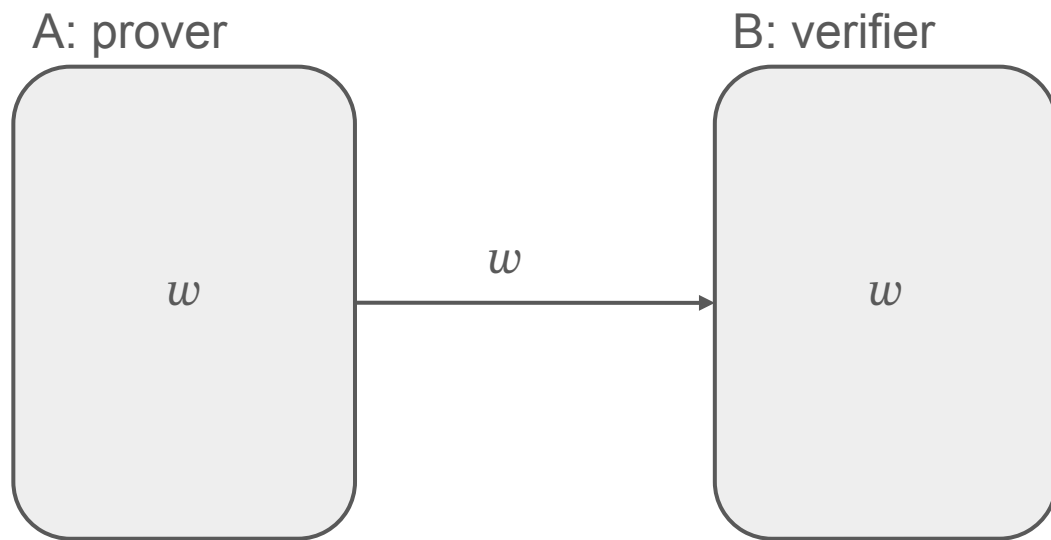
B: verifier



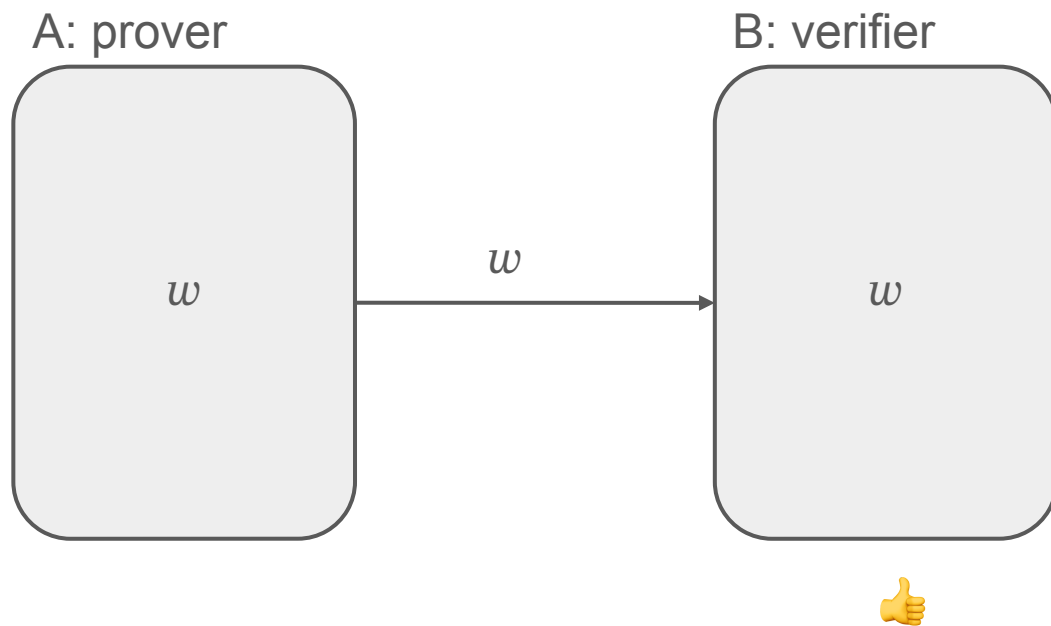
# Primer intento: enviar el witness



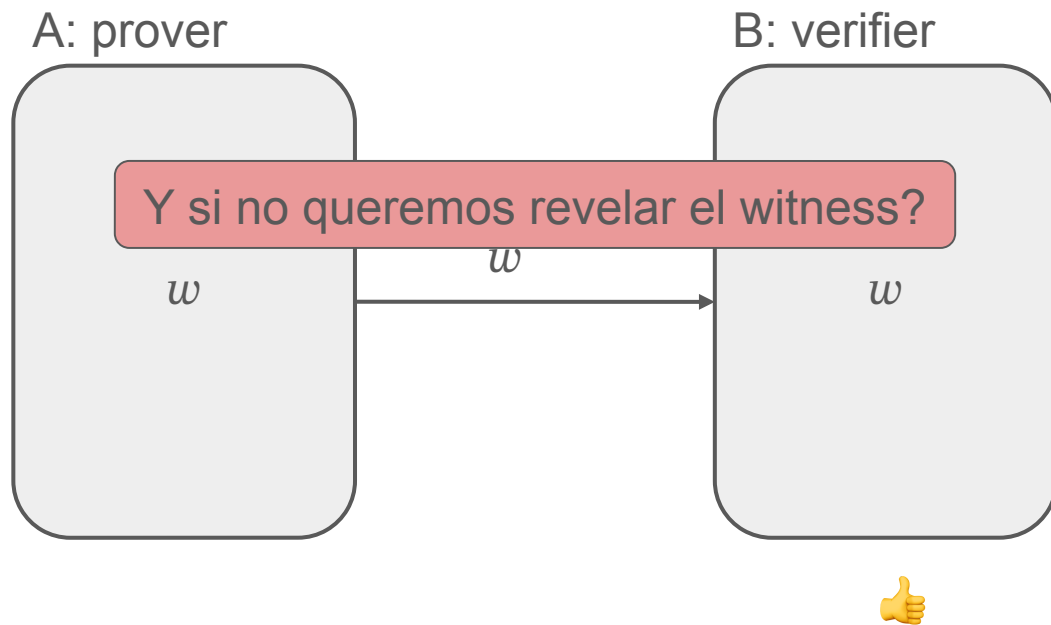
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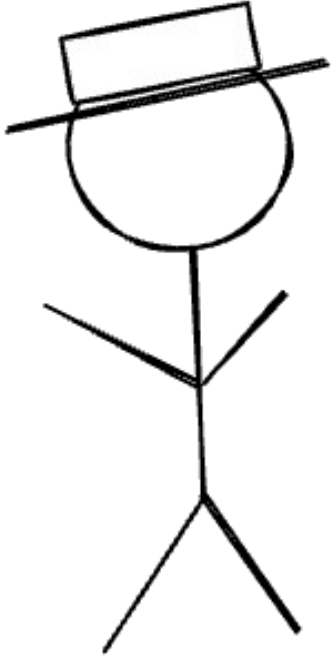


# Primer intento: enviar el witness

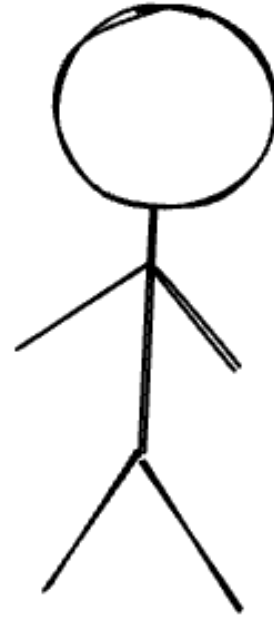


# Pruebas interactivas

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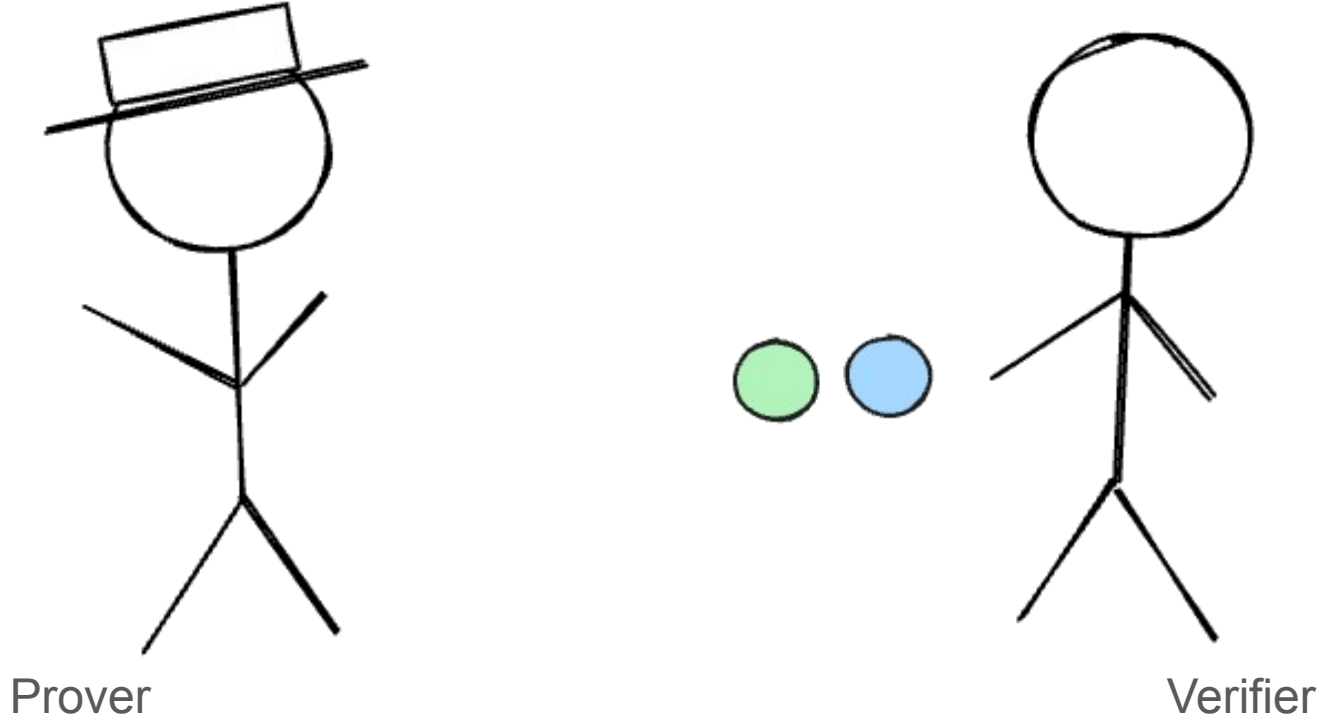
Prover



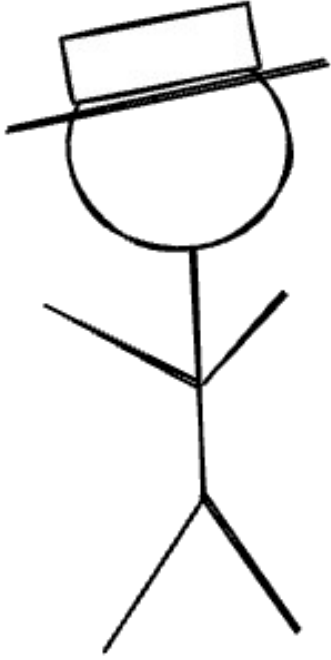
Verifier



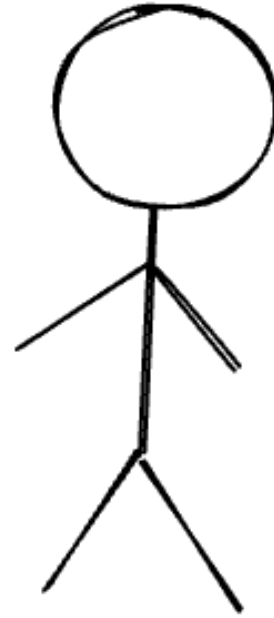
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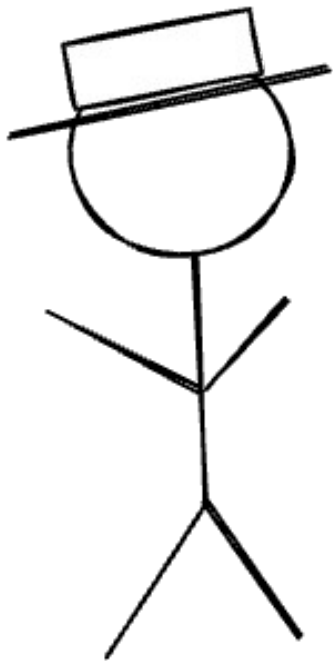


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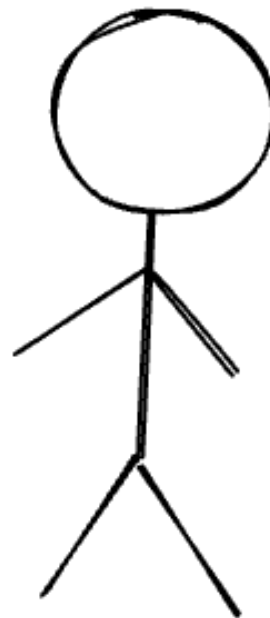


Verifier

# Pruebas interactivas



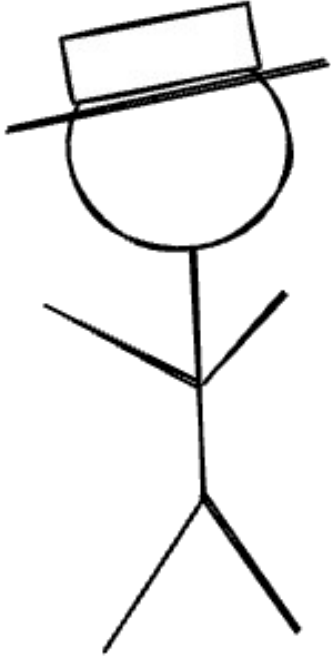
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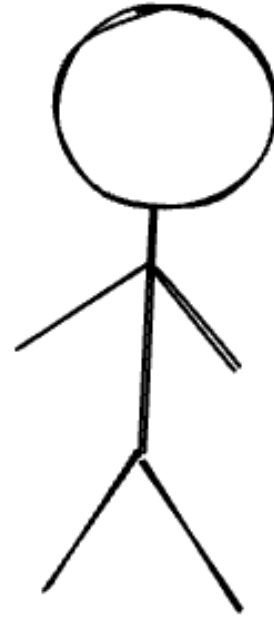
Verifier



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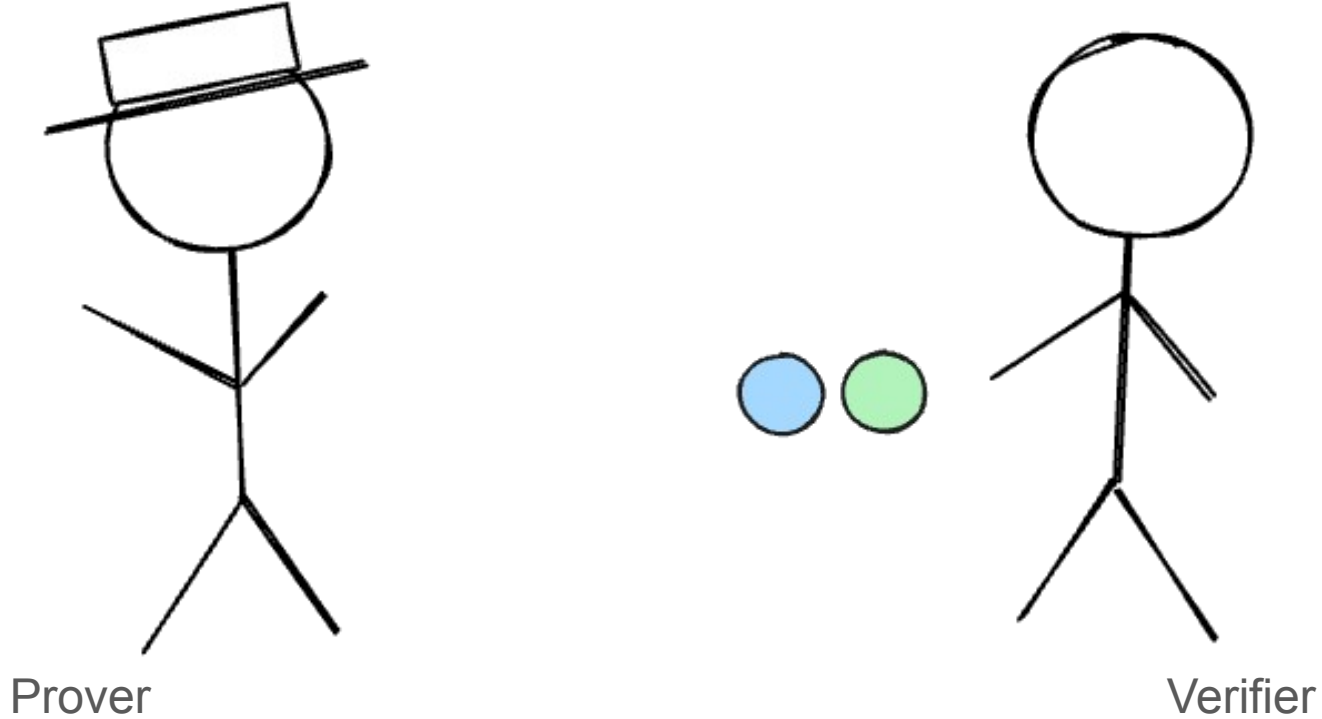


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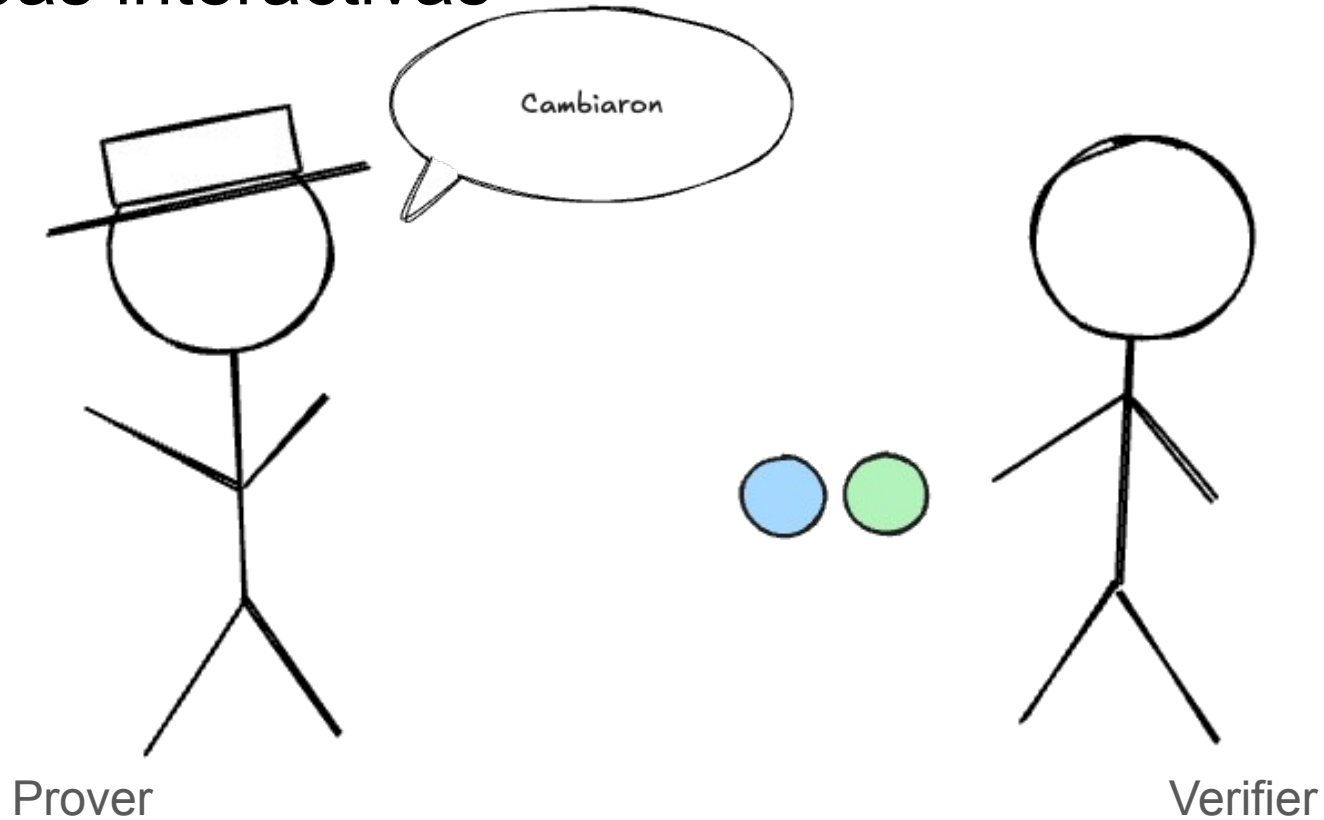


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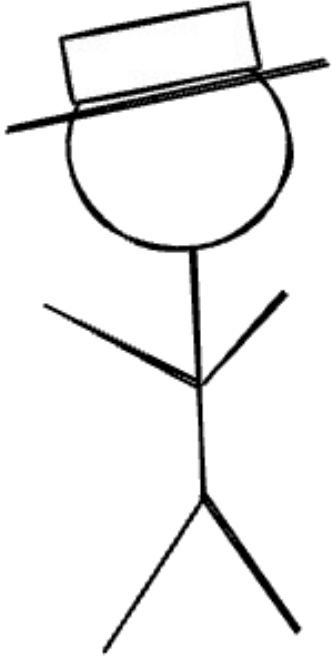
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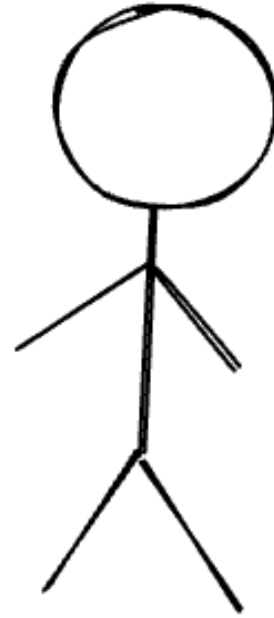
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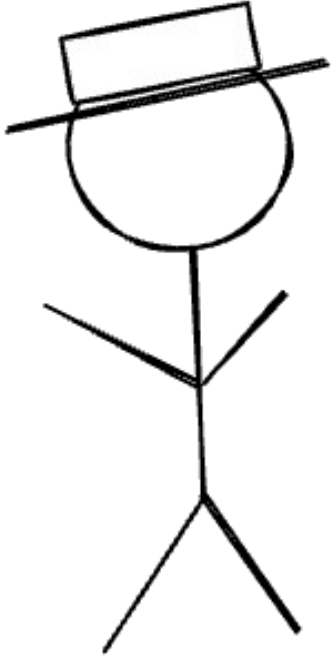


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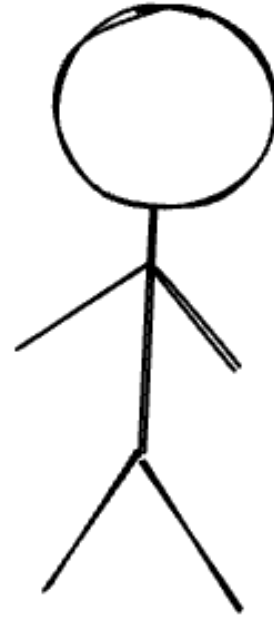


Verifier

# Pruebas interactivas



Prover

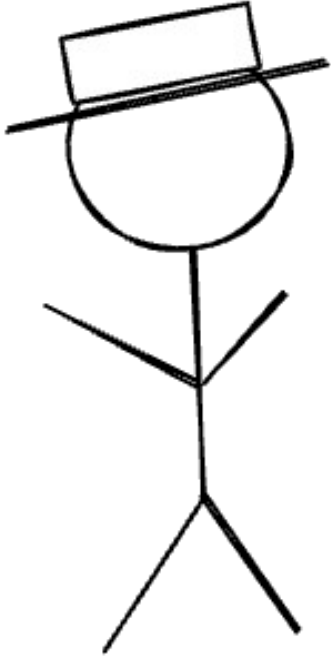


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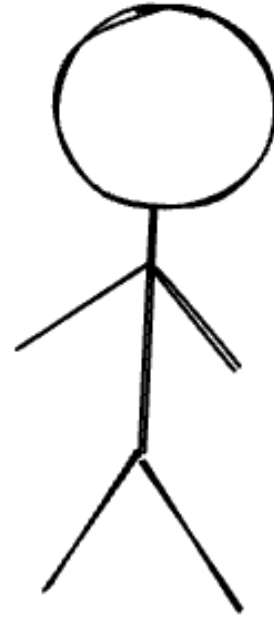




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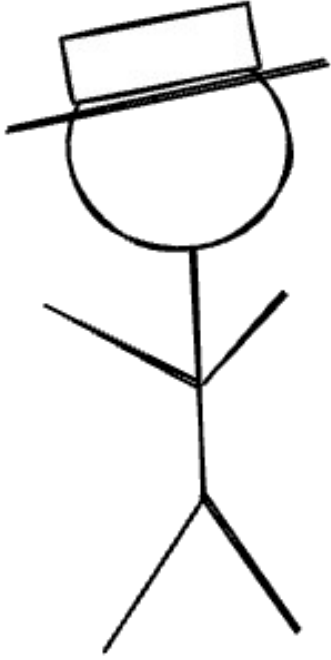


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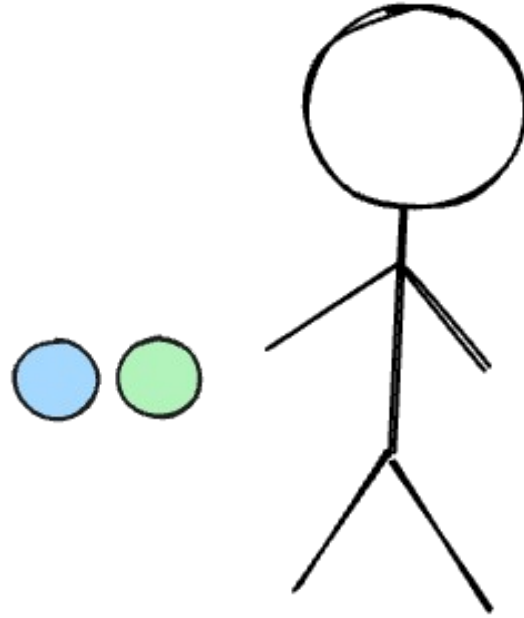


Verifier

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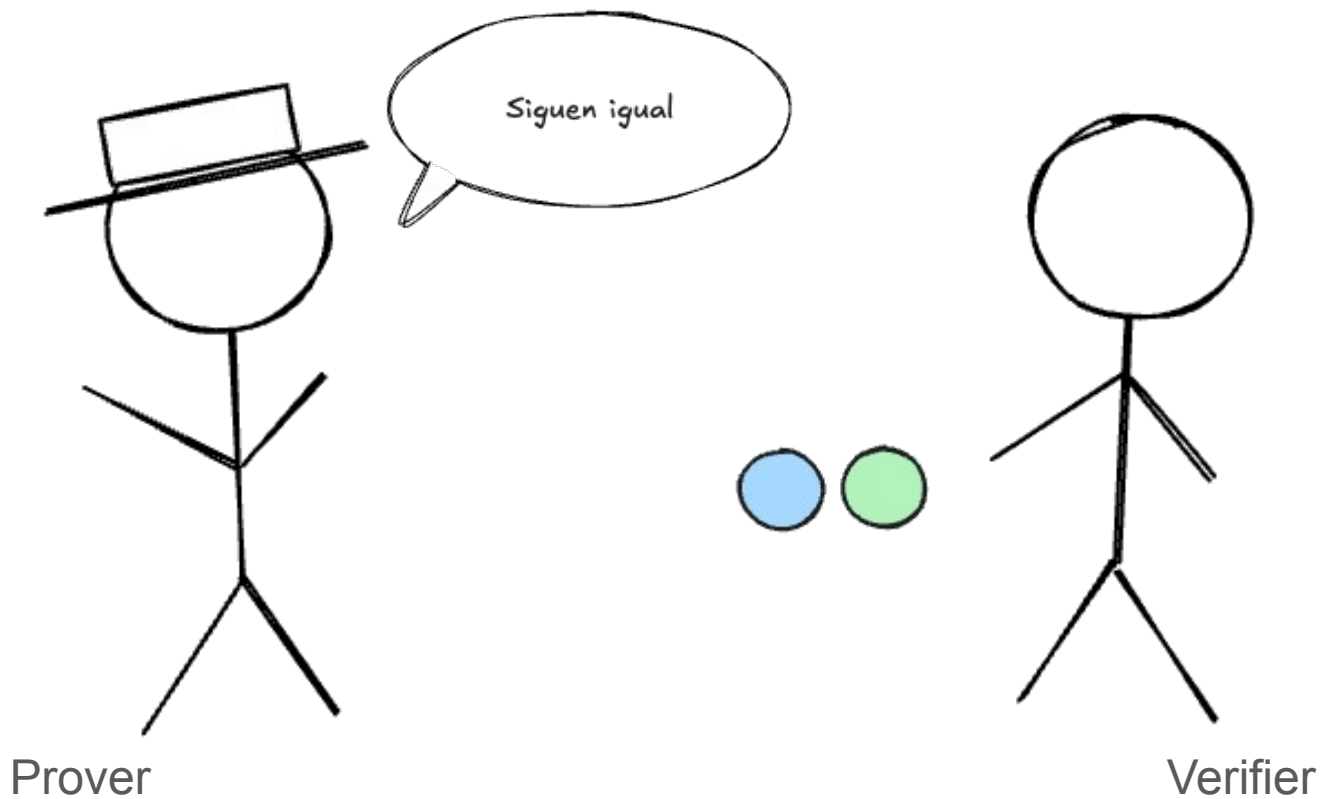


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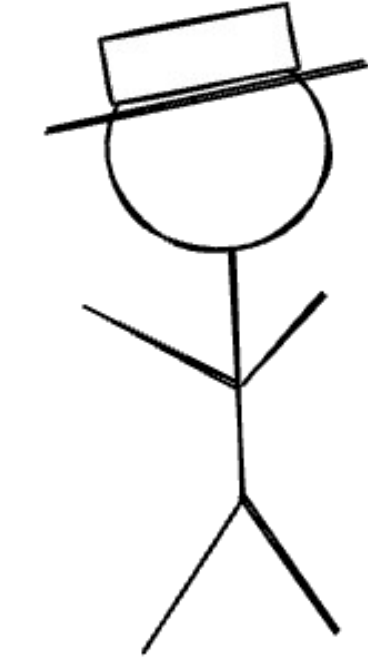


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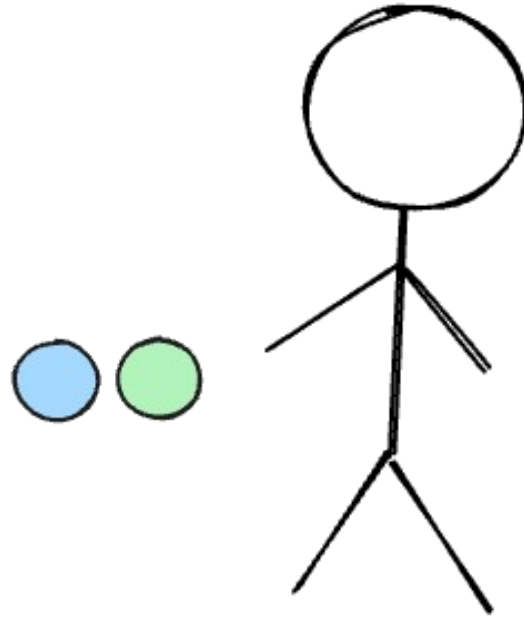
# Pruebas interactivas



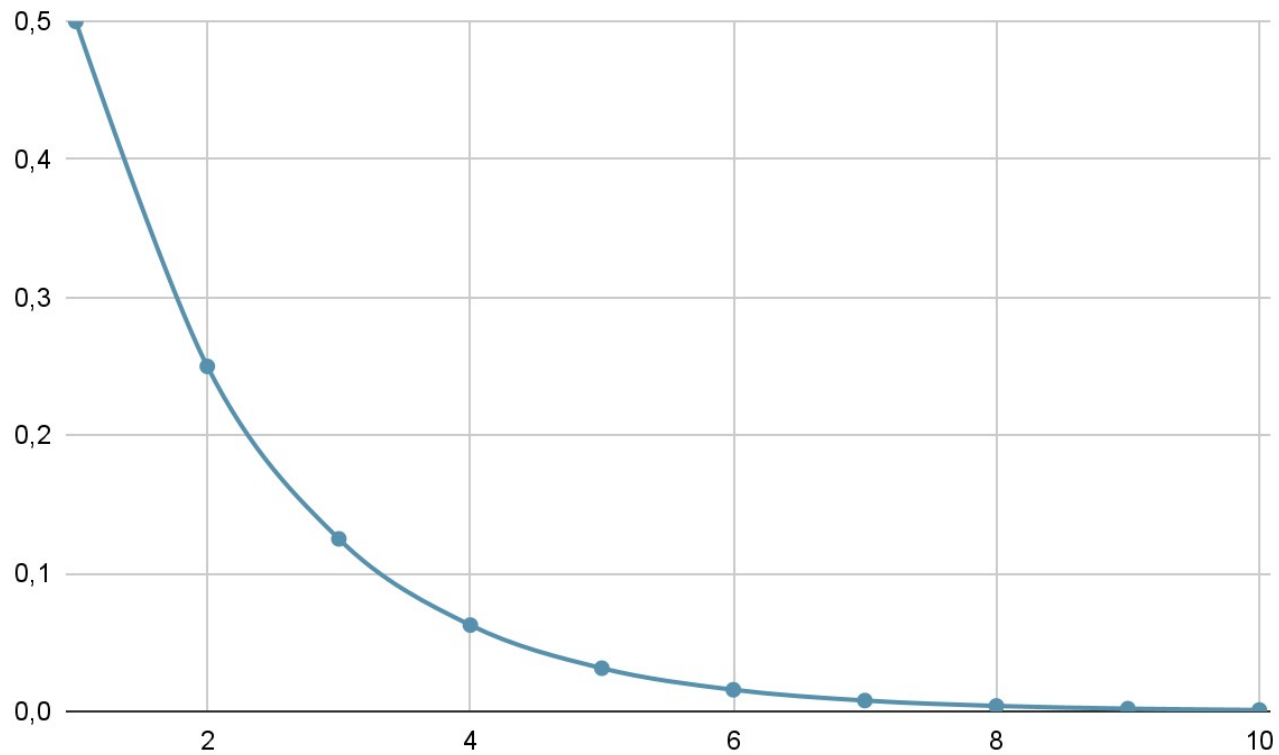
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Prover

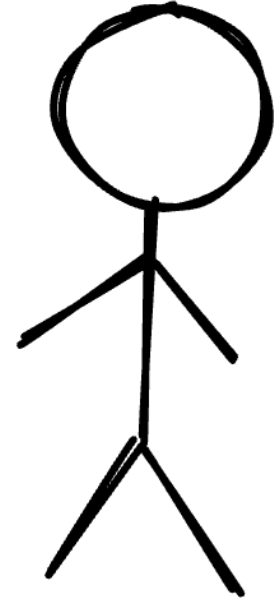
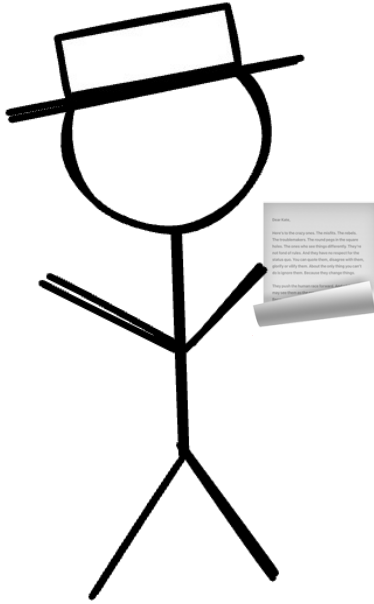


Verifier

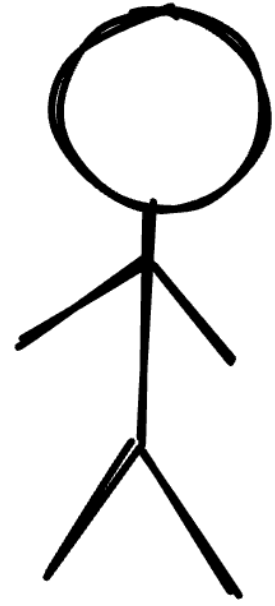
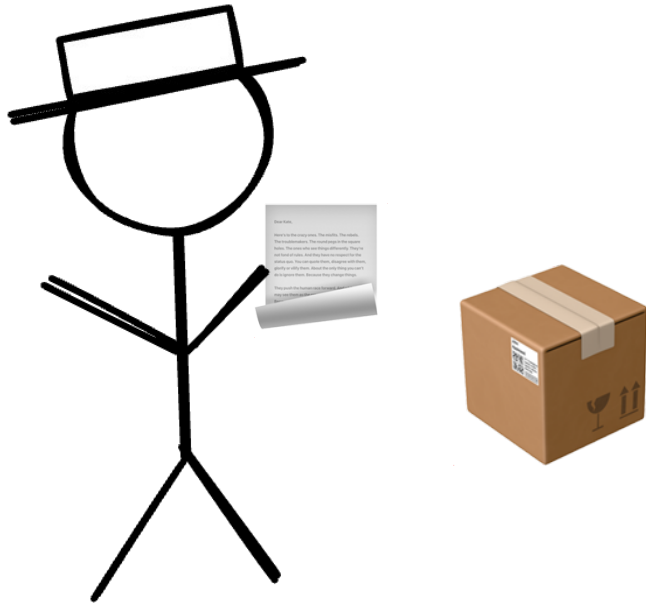


Probabilidad de “tener suerte” en  $k$  rondas seguidas

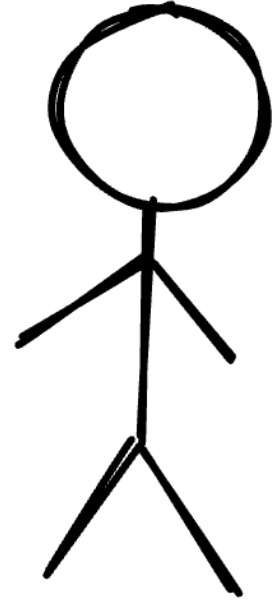
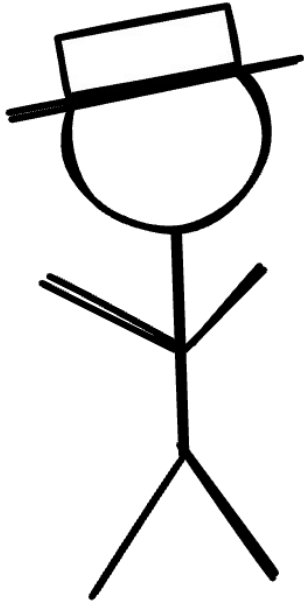
# Qué es un compromiso



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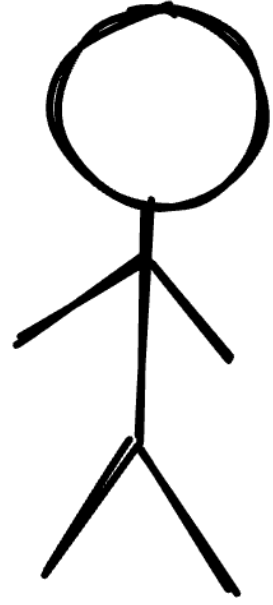
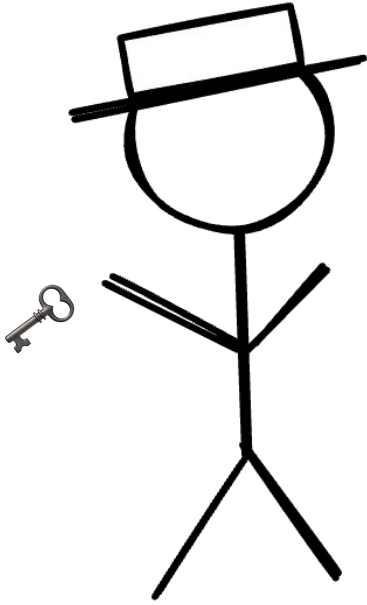


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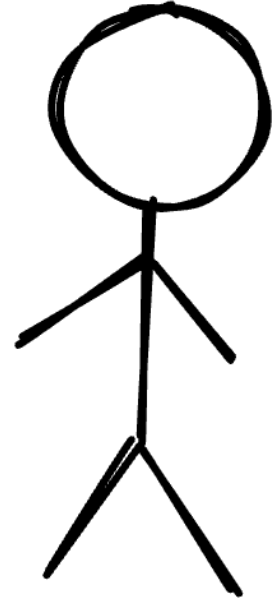
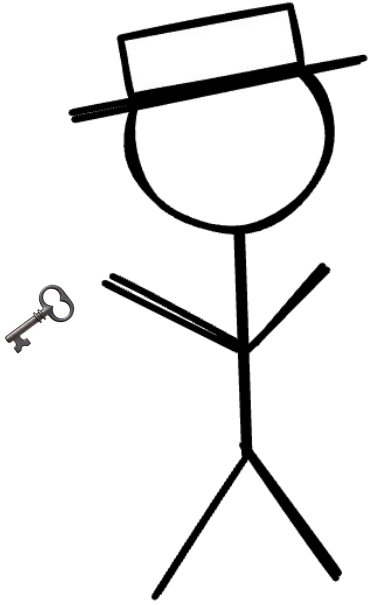




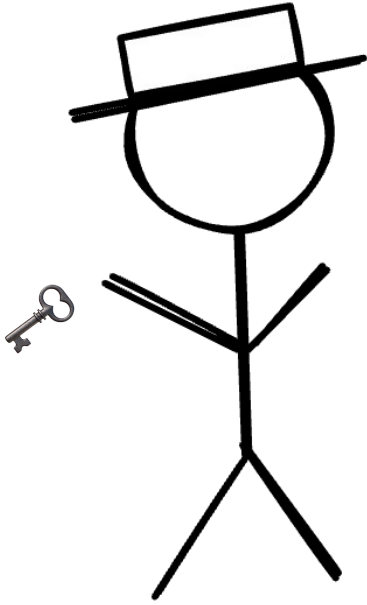
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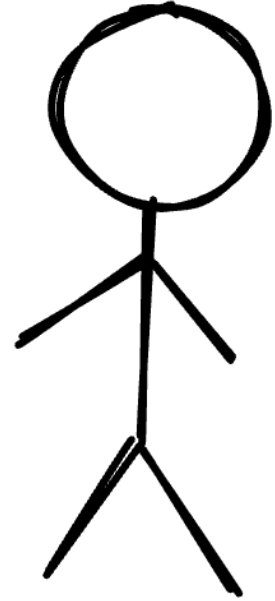
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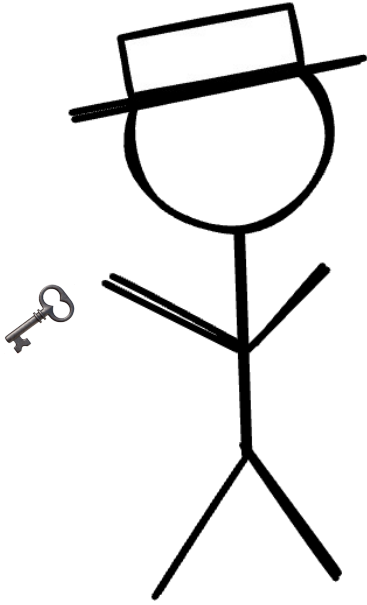
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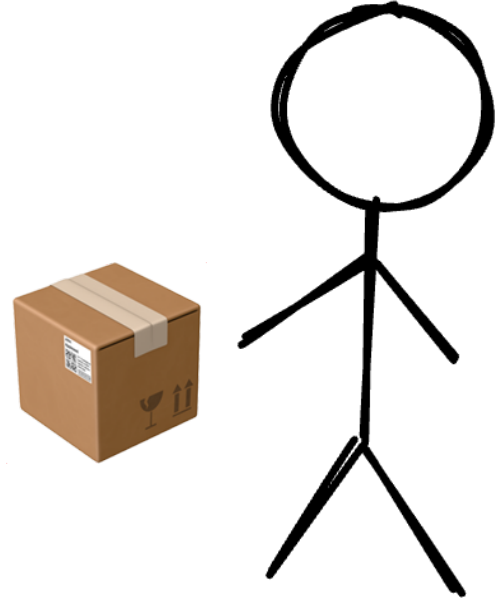
Hiding



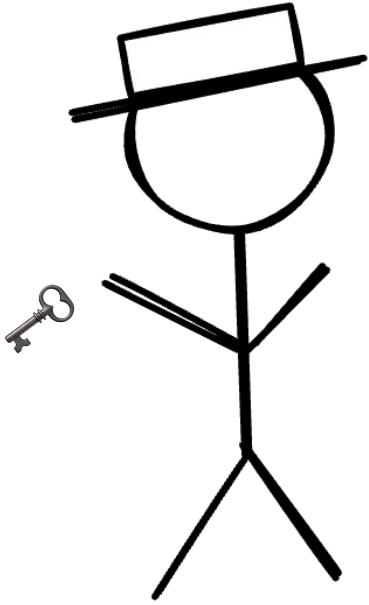
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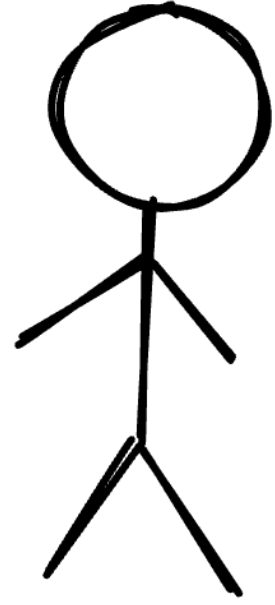
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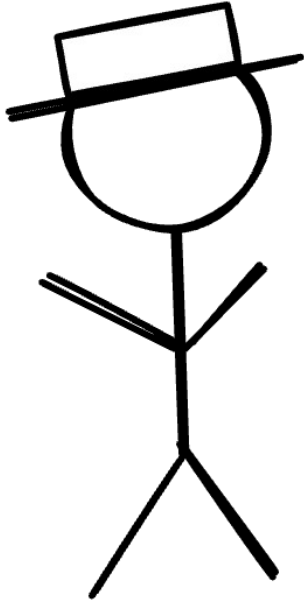
Hiding



Binding



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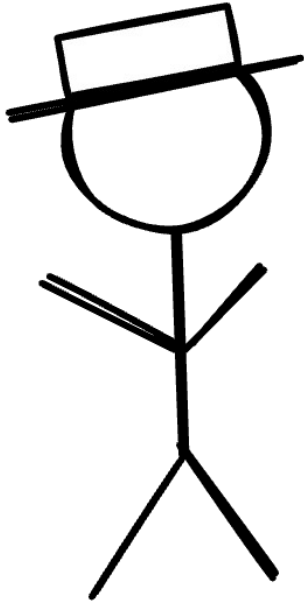


Hiding



Binding

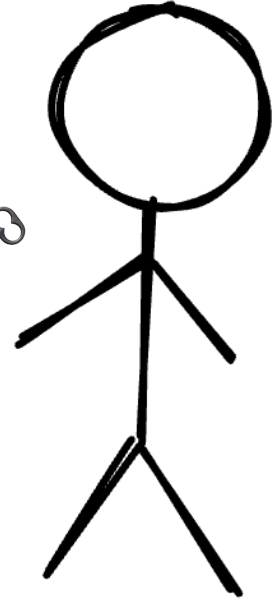
# Qué es un compromiso



Hiding



Binding



Apertura parcial de un compromiso



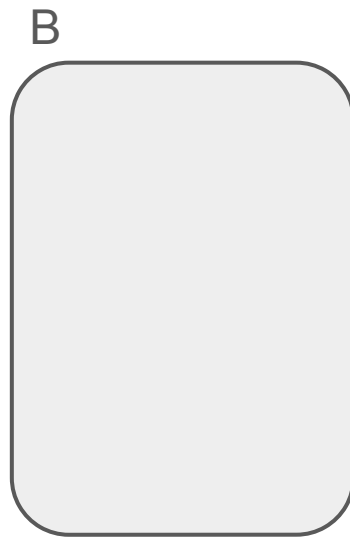
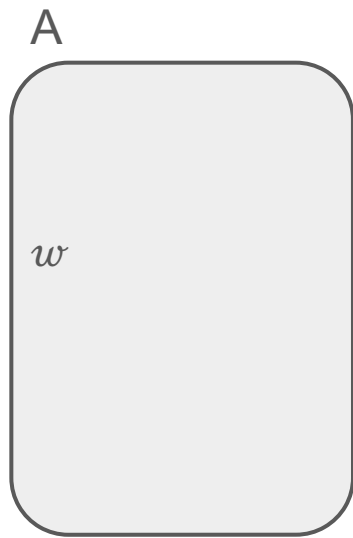
# Apertura parcial de un compromiso

- Permite revelar solo una parte del compromiso

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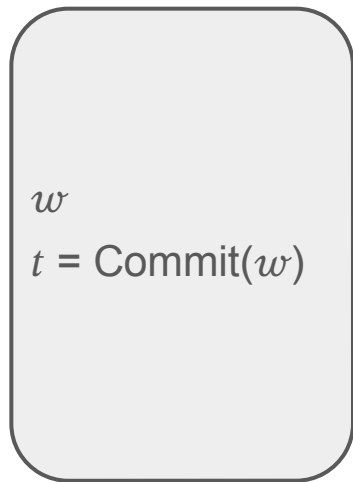
- Permite revelar solo una parte del compromiso
- Se pueden responder preguntas **verificables** sobre el compromiso

# Compromiso + desafíos



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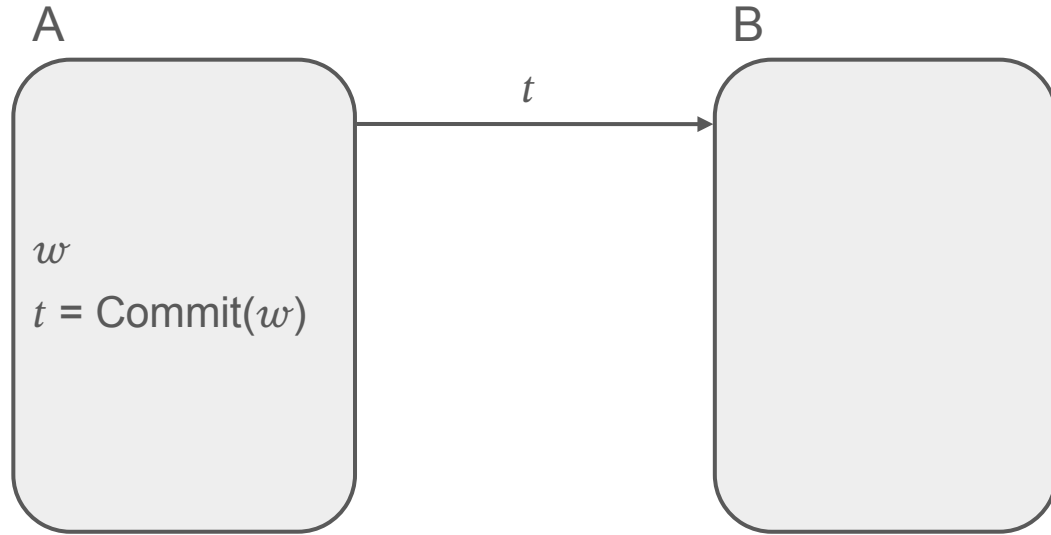
A



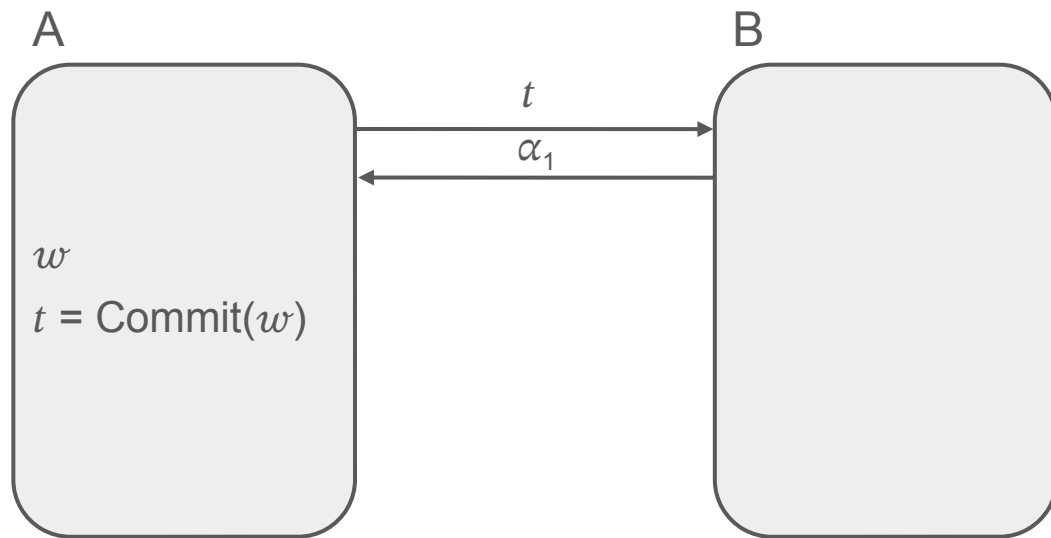
B



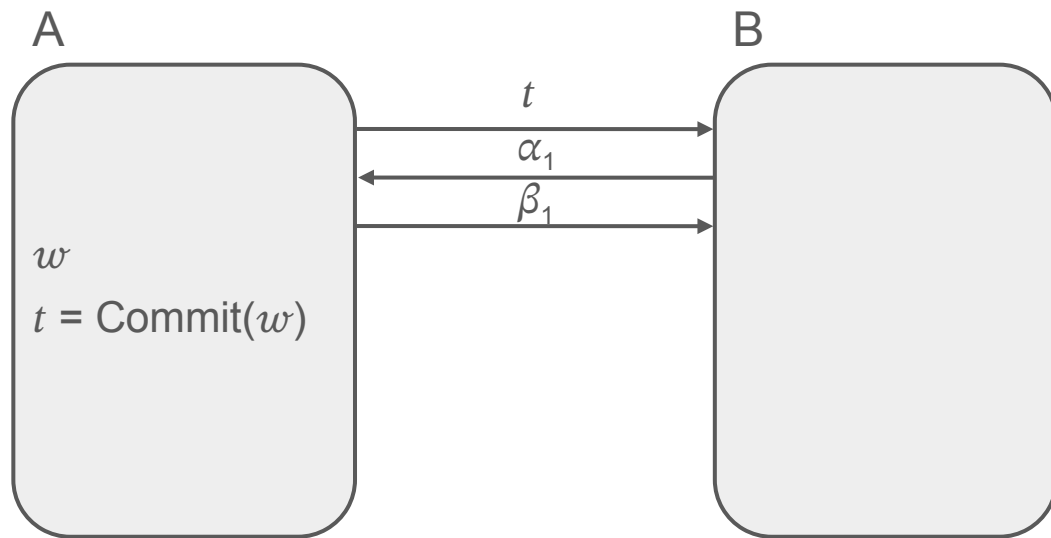
# Compromiso + desafíos



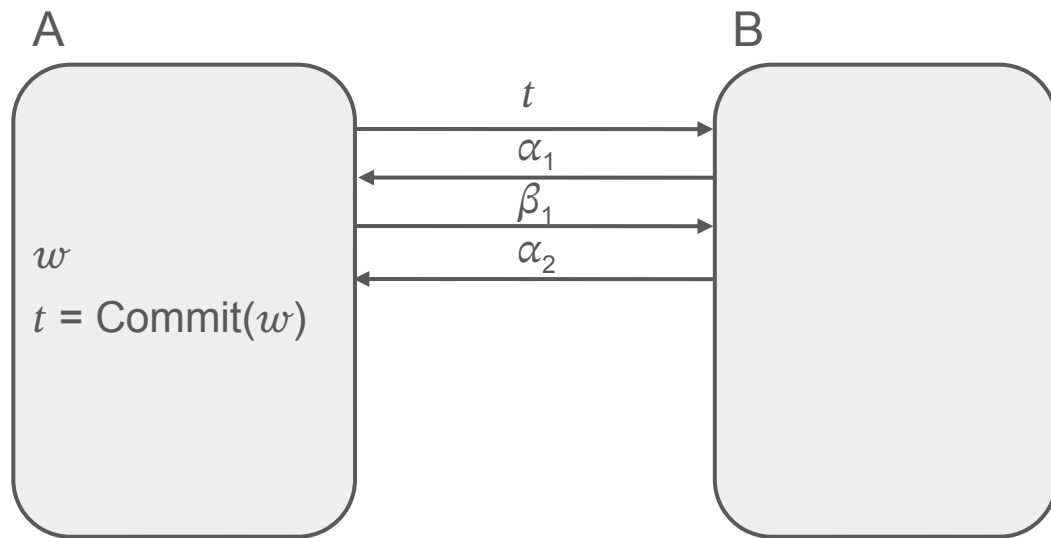
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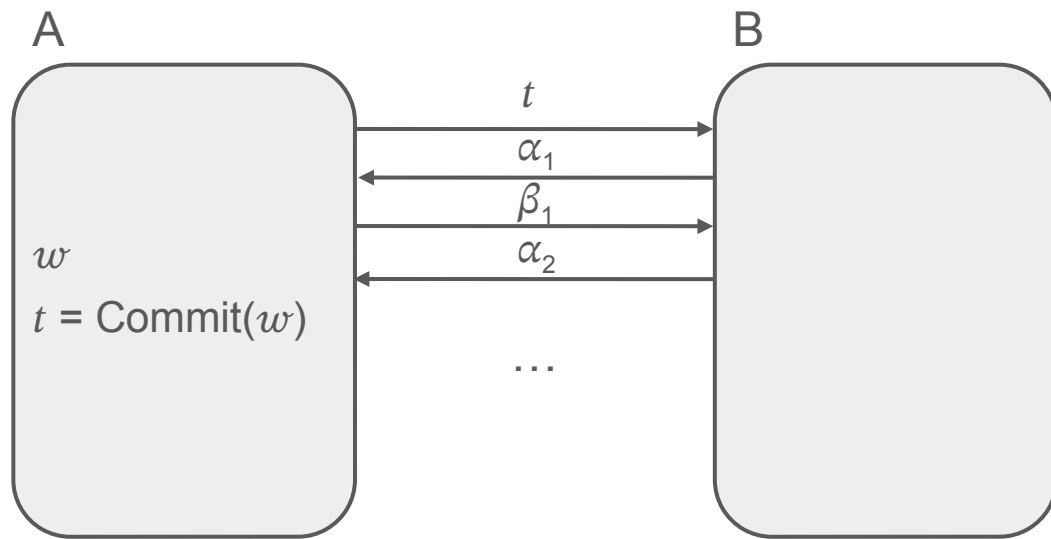


# Compromiso + desafíos

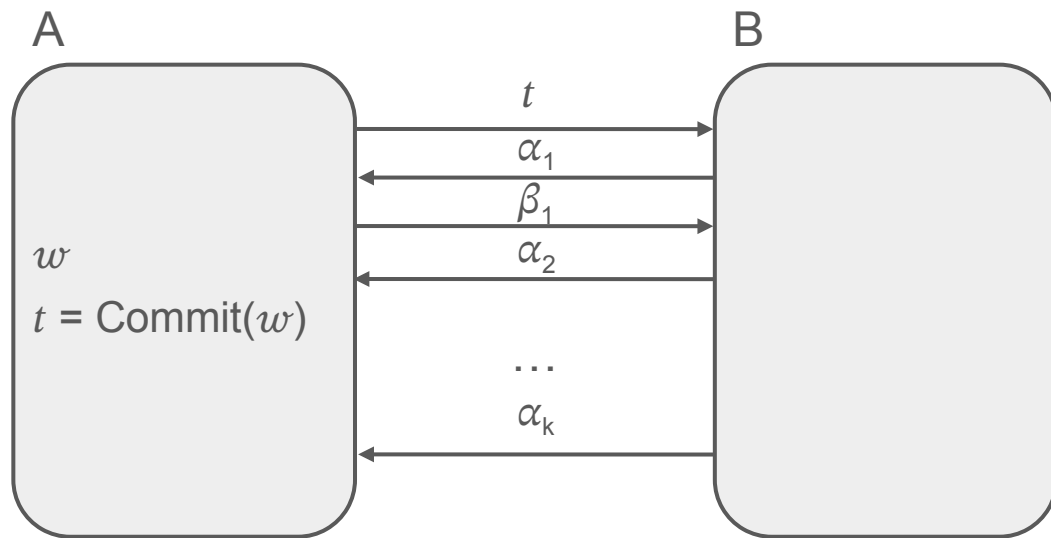




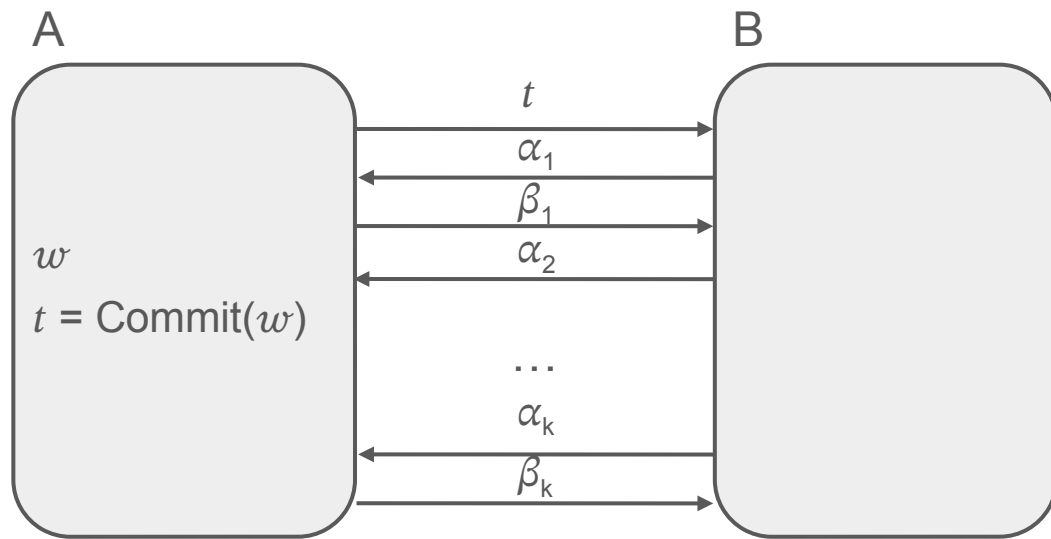
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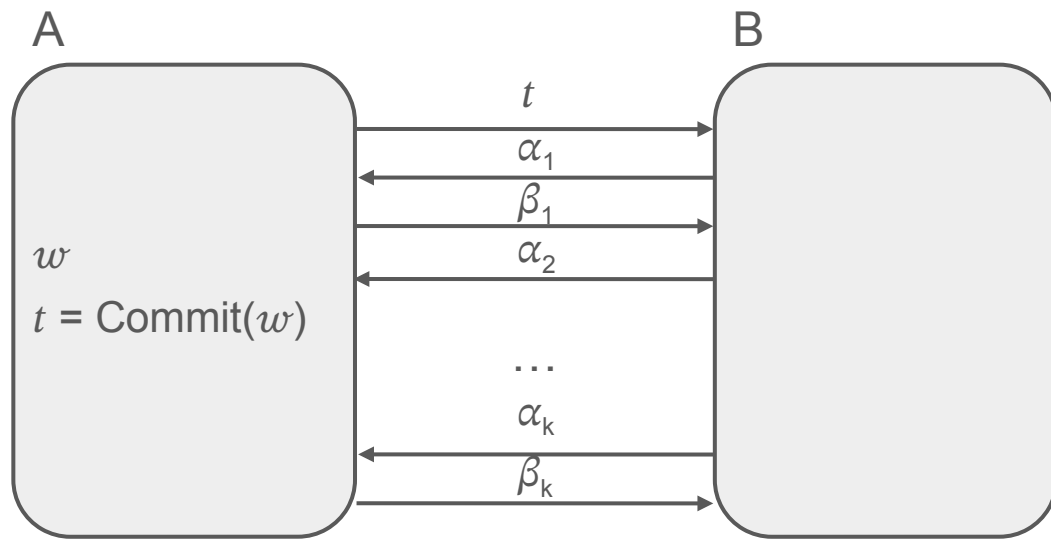
# Compromiso + desafíos



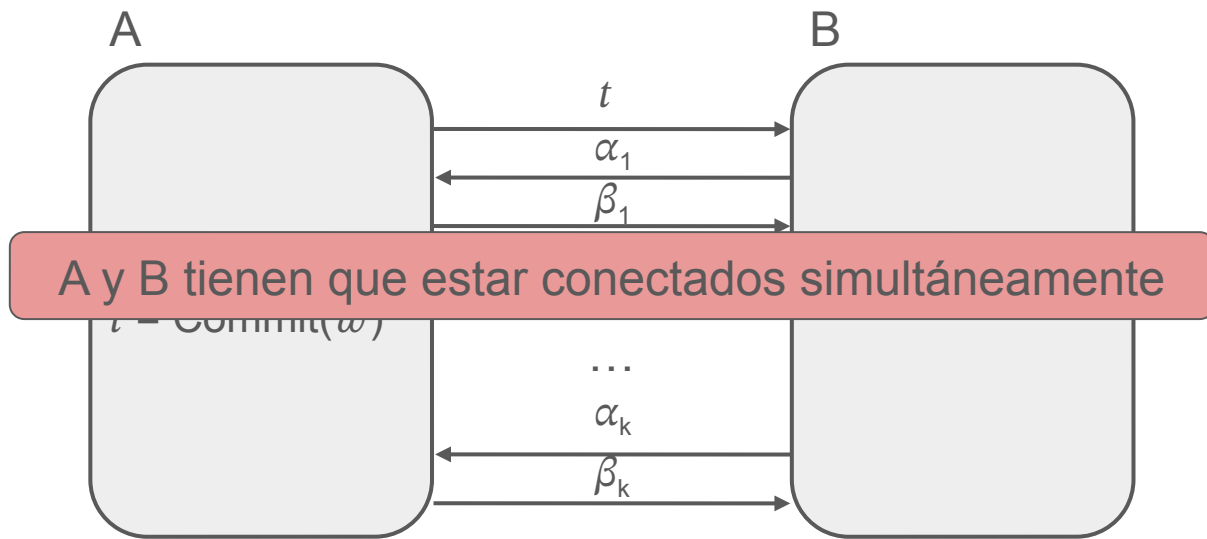
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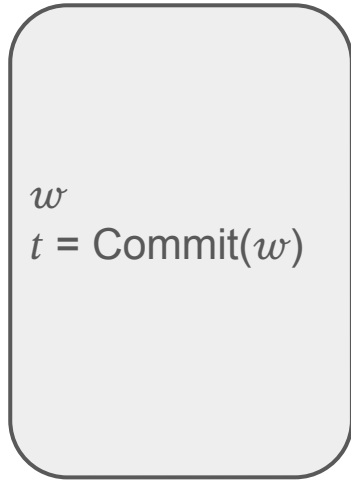
# Compromiso + desafíos



Pruebas no interactivas

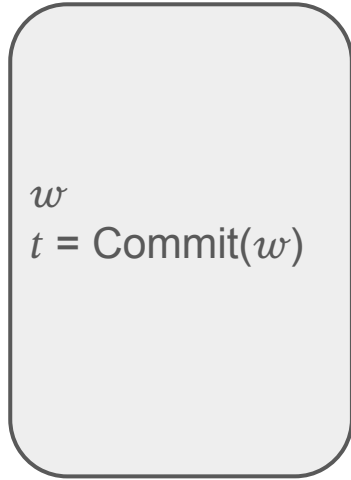
# De interactivo a no interactivo: Heurística de Fiat-Shamir

A

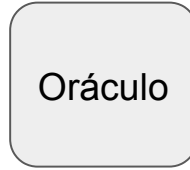


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A

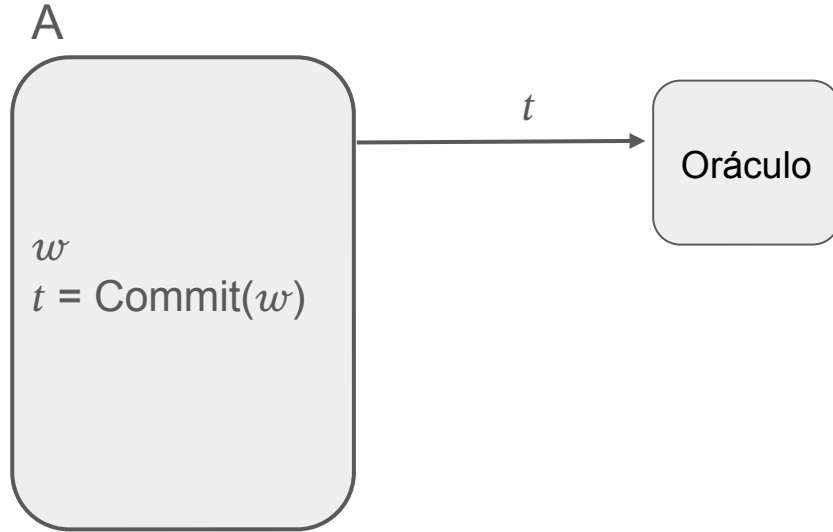


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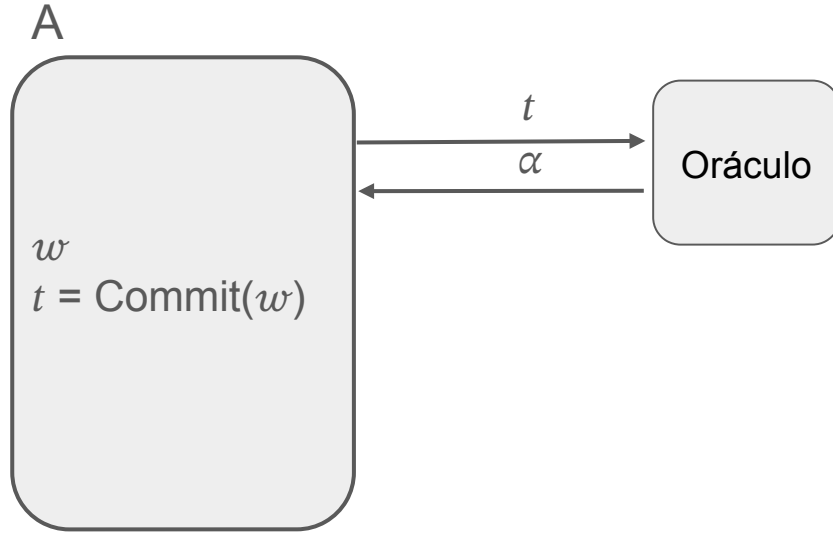




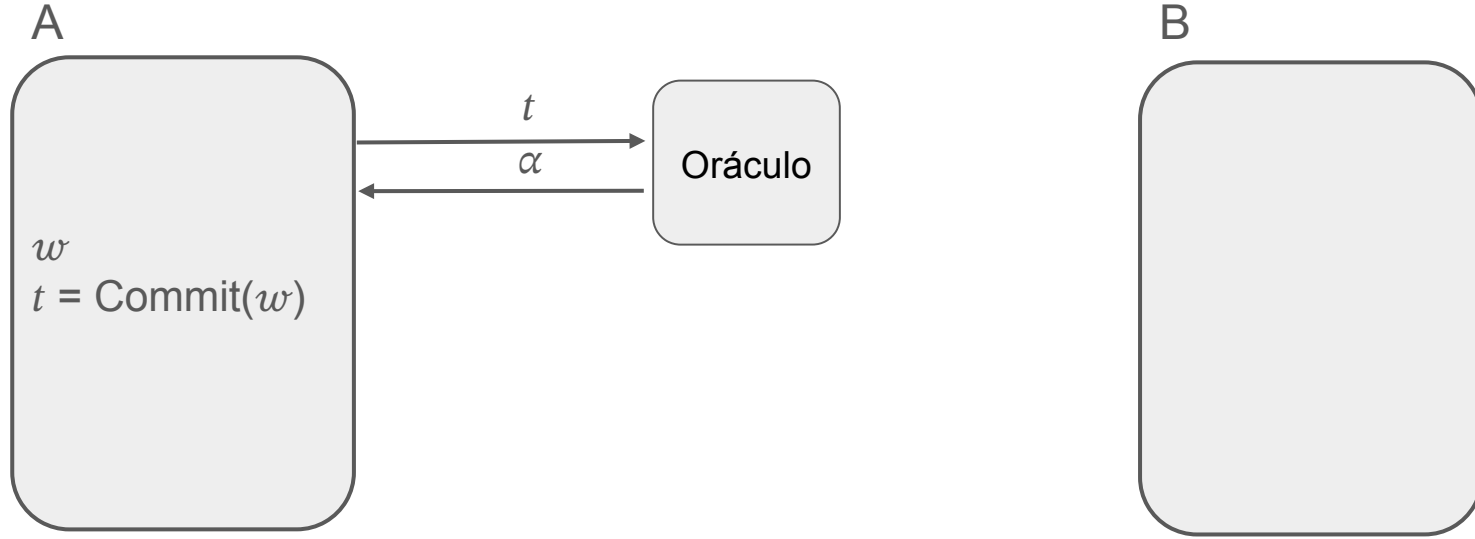
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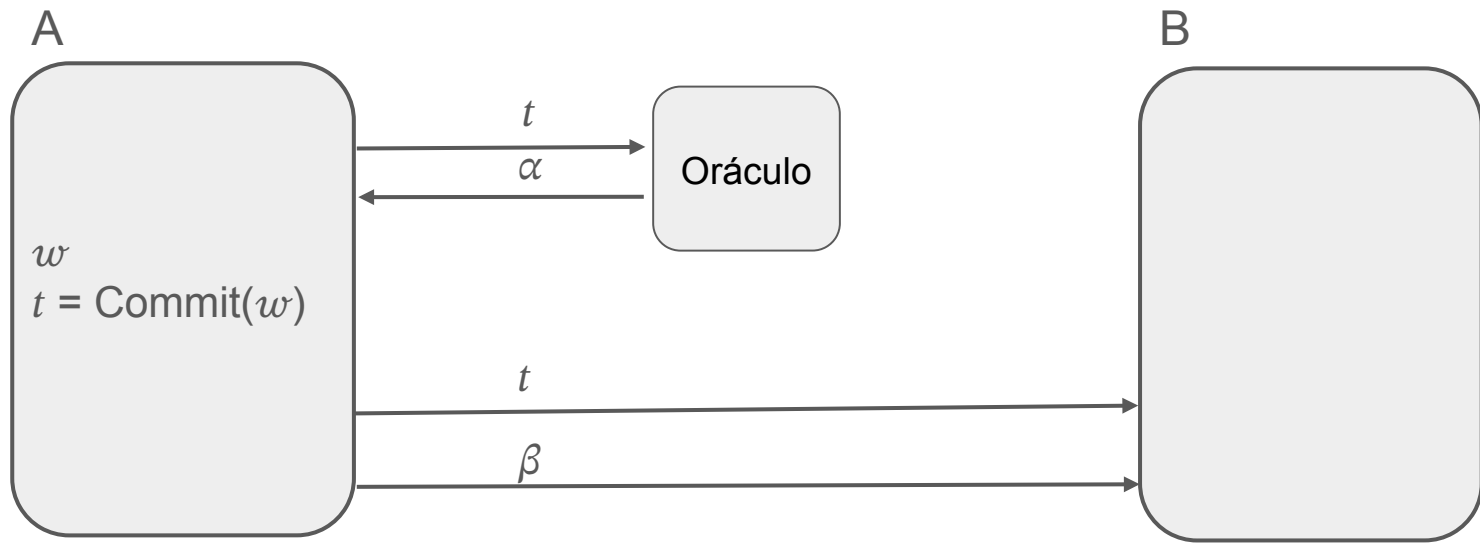
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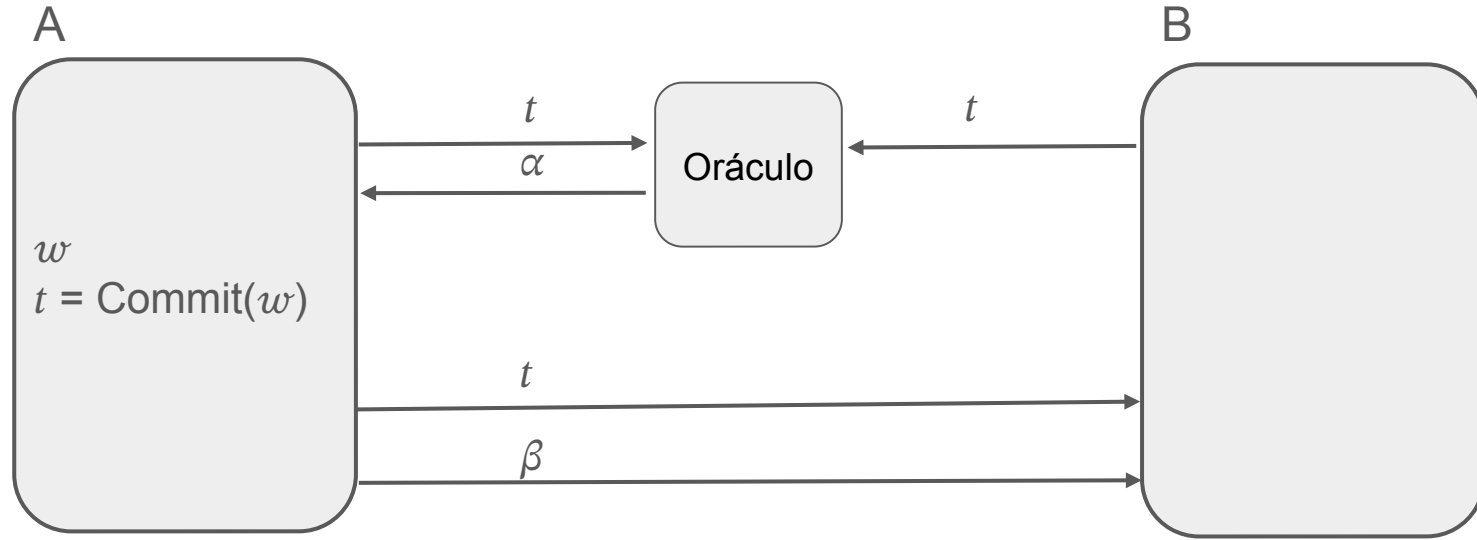
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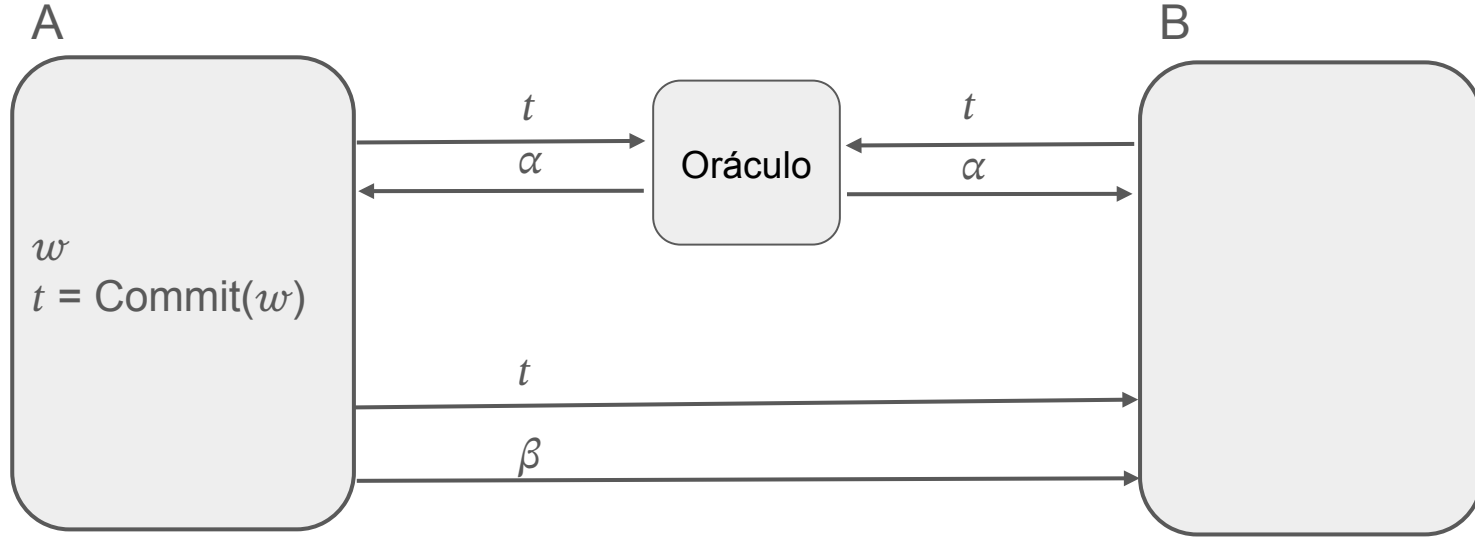
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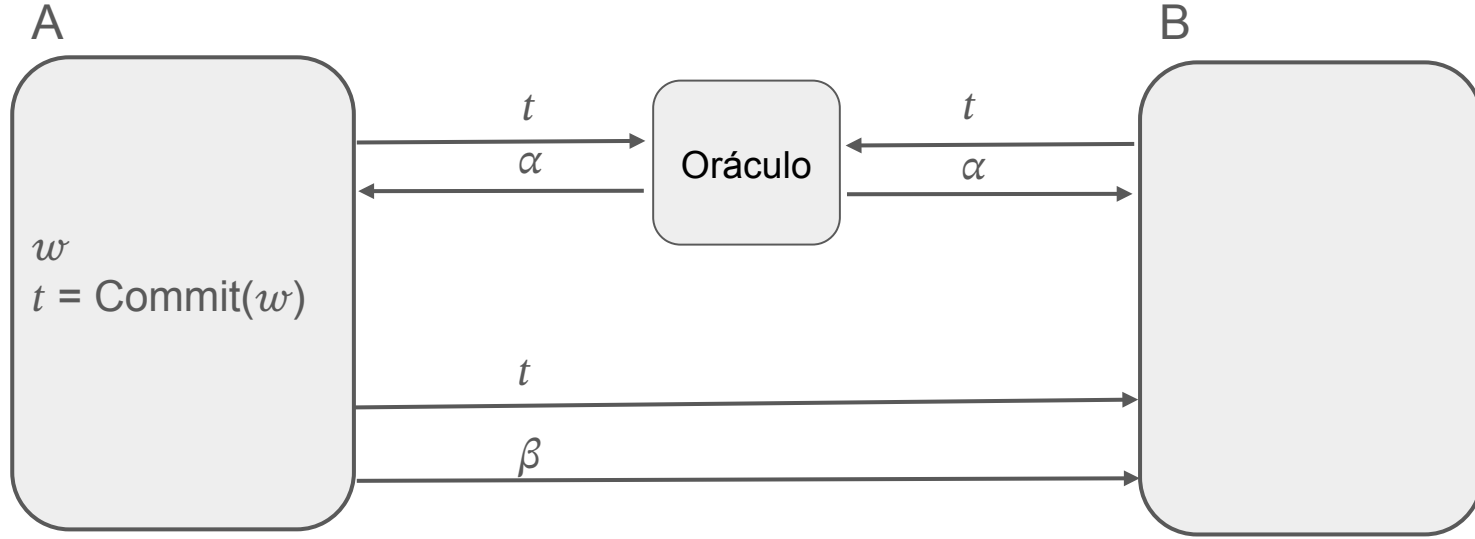
# De interactivo a no interactivo: Heurística de Fiat-Shamir



# De interactivo a no interactivo: Heurística de Fiat-Shamir

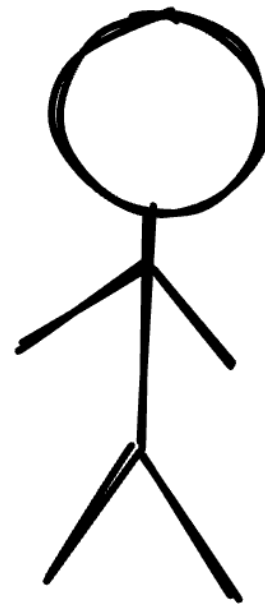
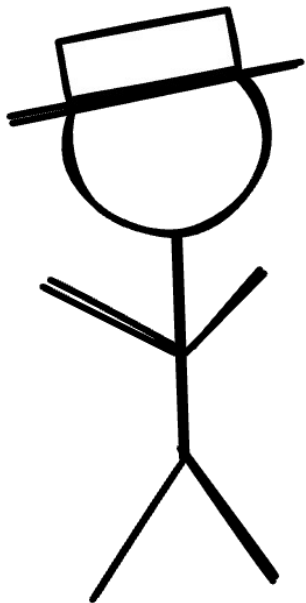


# De interactivo a no interactivo: Heurística de Fiat-Shamir



# Ejemplo: Sudoku 1/3

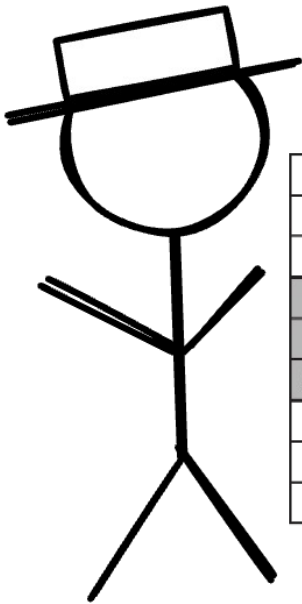
|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
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| 7 | 6 |   | 1 |  | 9 | 3 |   |   |
|   |   | 3 |   |  | 4 |   |   | 8 |
| 6 |   |   |   |  |   | 7 | 3 |   |
| 9 |   |   | 5 |  | 3 |   |   | 4 |
|   | 1 | 7 |   |  |   |   |   | 9 |
| 8 |   |   | 2 |  |   | 5 |   |   |
|   |   | 6 | 8 |  | 5 |   | 1 | 7 |
| 4 | 5 |   |   |  | 7 |   | 8 | 6 |



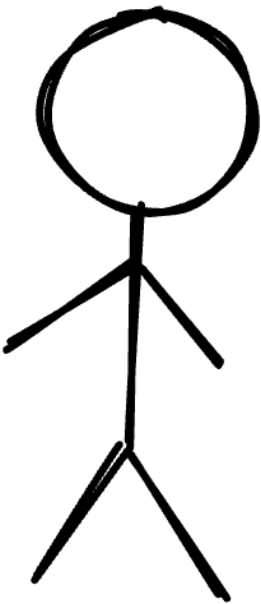


# Ejemplo: Sudoku 1/3

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
| 5 | 2 |   | 6 |  |   |   | 7 | 1 |
| 7 | 6 |   | 1 |  | 9 | 3 |   |   |
|   |   | 3 |   |  | 4 |   |   | 8 |
| 6 |   |   |   |  |   | 7 | 3 |   |
| 9 |   |   | 5 |  | 3 |   |   | 4 |
|   | 1 | 7 |   |  |   |   |   | 9 |
| 8 |   |   | 2 |  |   | 5 |   |   |
|   |   | 6 | 8 |  | 5 |   | 1 | 7 |
| 4 | 5 |   |   |  | 7 |   | 8 | 6 |

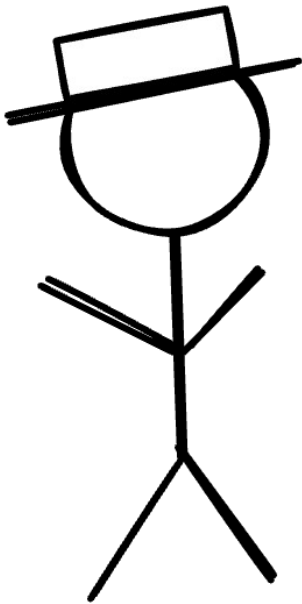


|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 | 4 | 6 | 3 | 8 | 9 | 7 | 1 |
| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
| 2 | 3 | 6 | 8 | 9 | 5 | 4 | 1 | 7 |
| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |

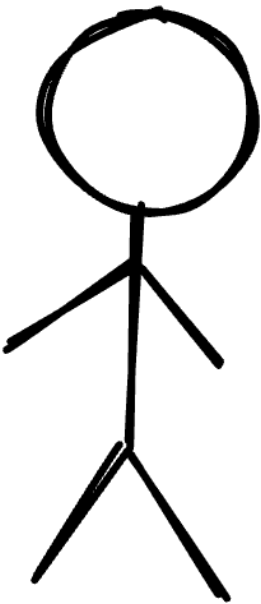


# Ejemplo: Sudoku 1/3

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
| 5 | 2 |   | 6 |  |   |   | 7 | 1 |
| 7 | 6 |   | 1 |  | 9 | 3 |   |   |
|   |   | 3 |   |  | 4 |   |   | 8 |
| 6 |   |   |   |  |   | 7 | 3 |   |
| 9 |   |   | 5 |  | 3 |   |   | 4 |
|   | 1 | 7 |   |  |   |   |   | 9 |
| 8 |   |   | 2 |  |   | 5 |   |   |
|   |   | 6 | 8 |  | 5 |   | 1 | 7 |
| 4 | 5 |   |   |  | 7 |   | 8 | 6 |

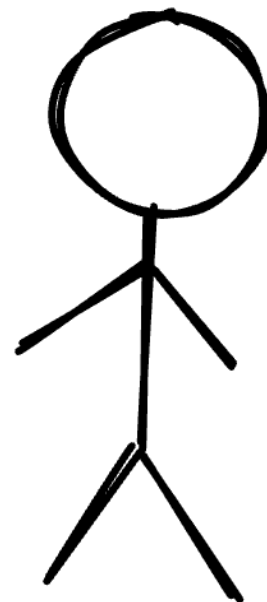
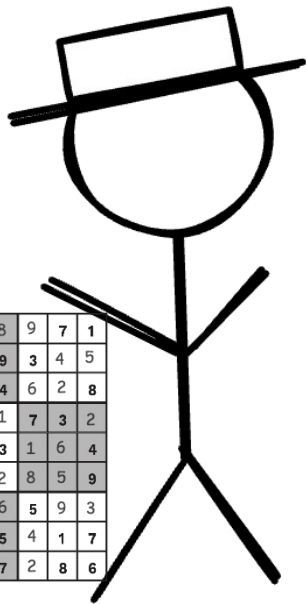


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|---|---|---|---|---|---|---|---|---|
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| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
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| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |



## Ejemplo: Sudoku 2/3

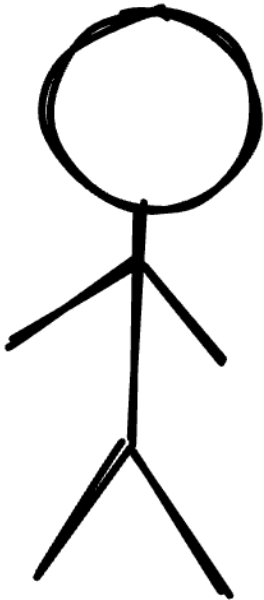
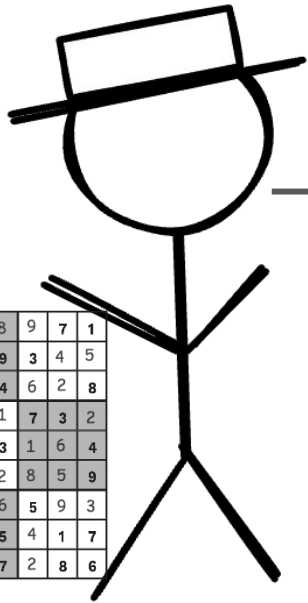
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| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
| 2 | 3 | 6 | 8 | 9 | 5 | 4 | 1 | 7 |
| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |



# Ejemplo: Sudoku 2/3

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
| 5 | 2 |   | 6 |  |   |   | 7 | 1 |
| 7 | 6 |   | 1 |  | 9 | 3 |   |   |
|   |   | 3 |   |  | 4 |   |   | 8 |
| 6 |   |   |   |  |   | 7 | 3 |   |
| 9 |   |   | 5 |  | 3 |   |   | 4 |
|   | 1 | 7 |   |  |   |   |   | 9 |
| 8 |   |   | 2 |  |   | 5 |   |   |
|   |   | 6 | 8 |  | 5 |   | 1 | 7 |
| 4 | 5 |   |   |  | 7 |   | 8 | 6 |

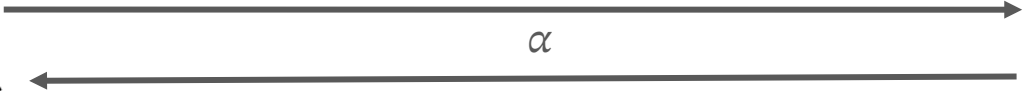
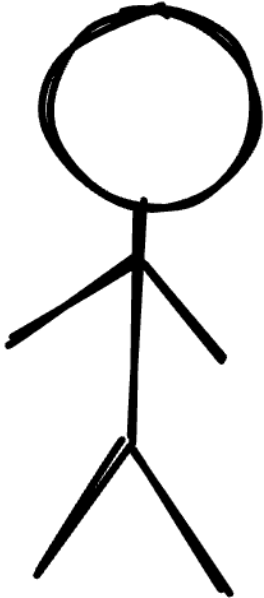
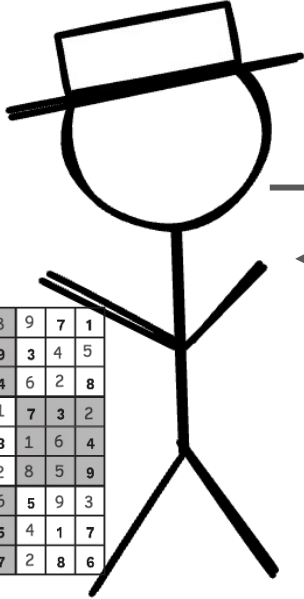
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| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
| 2 | 3 | 6 | 8 | 9 | 5 | 4 | 1 | 7 |
| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |



# Ejemplo: Sudoku 2/3

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
| 5 | 2 |   | 6 |  |   |   | 7 | 1 |
| 7 | 6 |   | 1 |  | 9 | 3 |   |   |
|   |   | 3 |   |  | 4 |   |   | 8 |
| 6 |   |   |   |  |   | 7 | 3 |   |
| 9 |   |   | 5 |  | 3 |   |   | 4 |
|   | 1 | 7 |   |  |   |   |   | 9 |
| 8 |   |   | 2 |  |   | 5 |   |   |
|   |   | 6 | 8 |  | 5 |   | 1 | 7 |
| 4 | 5 |   |   |  | 7 |   | 8 | 6 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 | 4 | 6 | 3 | 8 | 9 | 7 | 1 |
| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
| 2 | 3 | 6 | 8 | 9 | 5 | 4 | 1 | 7 |
| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |



# Ejemplo: Sudoku 2/3

|   |   |   |   |  |   |   |   |   |
|---|---|---|---|--|---|---|---|---|
| 5 | 2 |   | 6 |  |   |   | 7 | 1 |
| 7 | 6 |   | 1 |  | 9 | 3 |   |   |
|   |   | 3 |   |  | 4 |   |   | 8 |
| 6 |   |   |   |  |   | 7 | 3 |   |
| 9 |   |   | 5 |  | 3 |   |   | 4 |
|   | 1 | 7 |   |  |   |   |   | 9 |
| 8 |   |   | 2 |  |   | 5 |   |   |
|   |   | 6 | 8 |  | 5 |   | 1 | 7 |
| 4 | 5 |   |   |  | 7 |   | 8 | 6 |

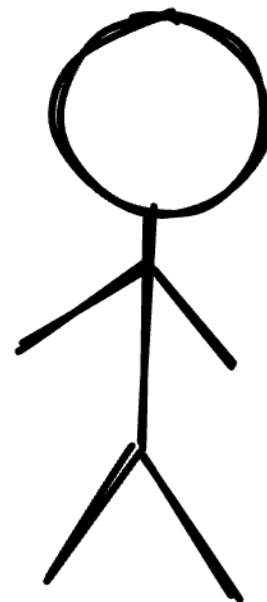
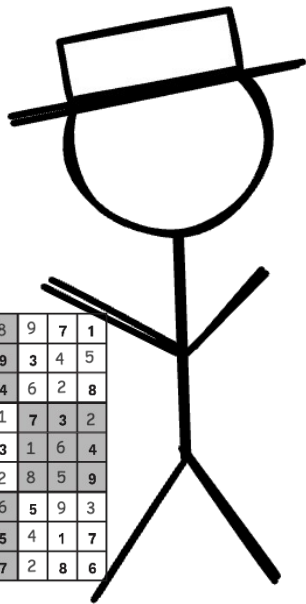
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|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 | 4 | 6 | 3 | 8 | 9 | 7 | 1 |
| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
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| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
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| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
| 2 | 3 | 6 | 8 | 9 | 5 | 4 | 1 | 7 |
| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 |   | 6 |   |   |   | 7 | 1 |
| 7 | 6 |   | 1 |   | 9 | 3 |   |   |
|   |   | 3 |   |   | 4 |   |   | 8 |
| 6 |   |   |   |   |   | 7 | 3 |   |
| 9 |   |   | 5 |   | 3 |   |   | 4 |
|   | 1 | 7 |   | 6 |   |   |   | 9 |
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|   |   | 6 | 8 |   | 5 |   | 1 | 7 |
| 4 | 5 |   |   |   | 7 |   | 8 | 6 |

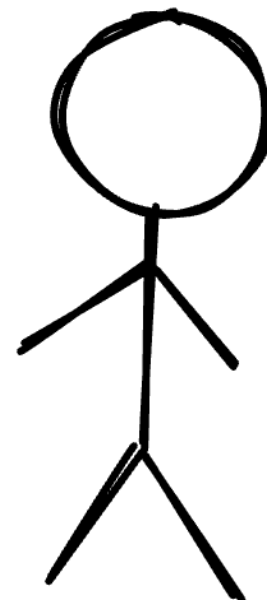
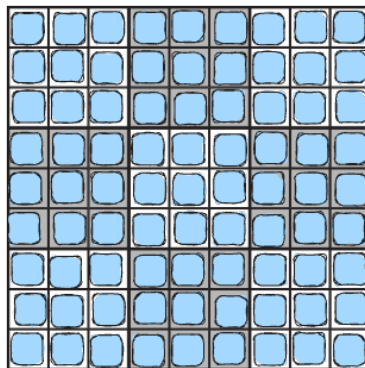
## Ejemplo: Sudoku 3/3

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 | 4 | 6 | 3 | 8 | 9 | 7 | 1 |
| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
| 2 | 3 | 6 | 8 | 9 | 5 | 4 | 1 | 7 |
| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |



# Ejemplo: Sudoku 3/3

Compromiso al  
tablero permutado

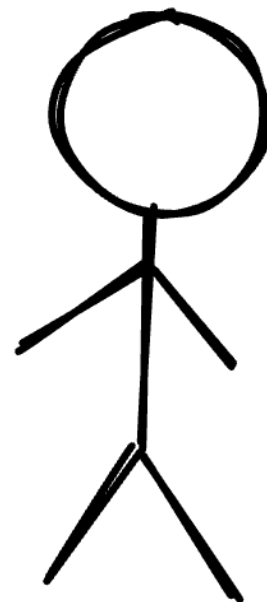
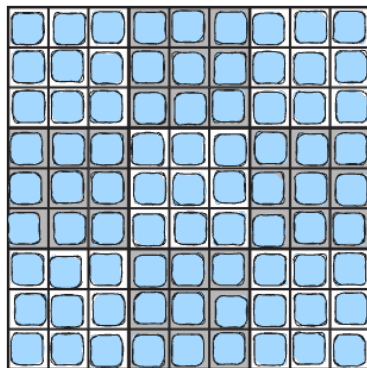


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| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
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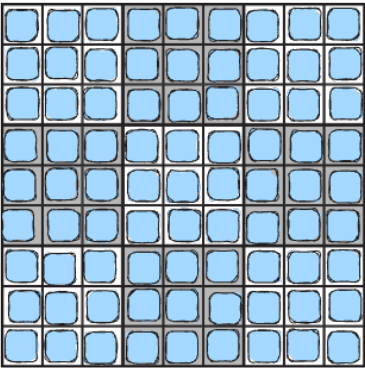
# Ejemplo: Sudoku 3/3

Compromiso al  
tablero permutado



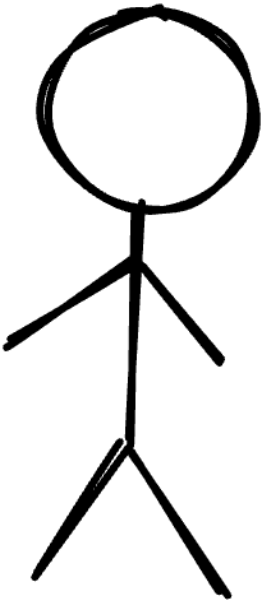
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| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
| 2 | 3 | 6 | 8 | 9 | 5 | 4 | 1 | 7 |
| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |

# Ejemplo: Sudoku 3/3



Compromiso al  
tablero permutado

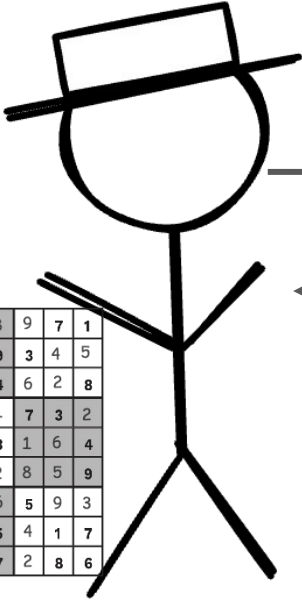
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| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
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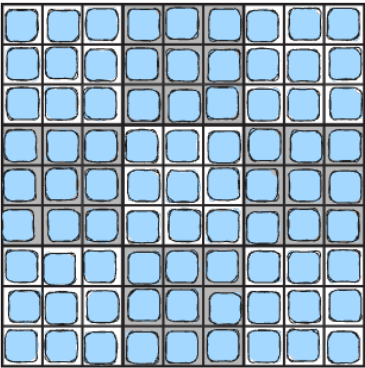
Fila  
Columna  
Bloque  
Permutación

# Ejemplo: Sudoku 3/3

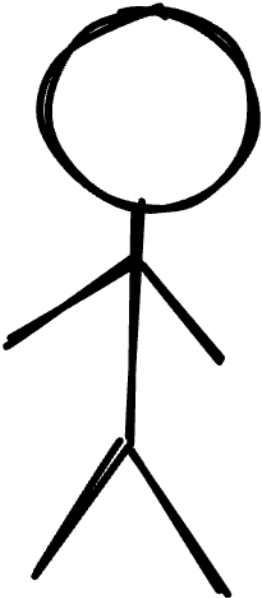
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| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
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Compromiso al  
tablero permutado



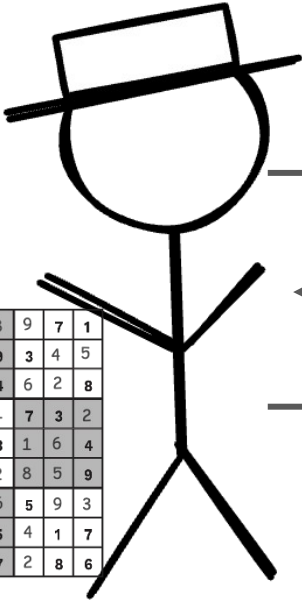
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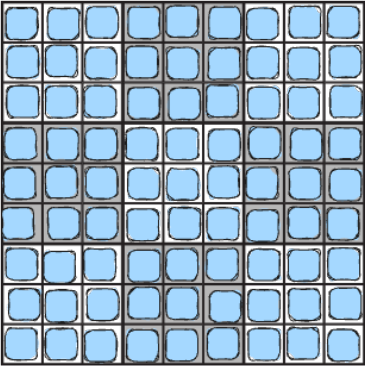
Fila  
Columna  
Bloque  
Permutación

# Ejemplo: Sudoku 3/3

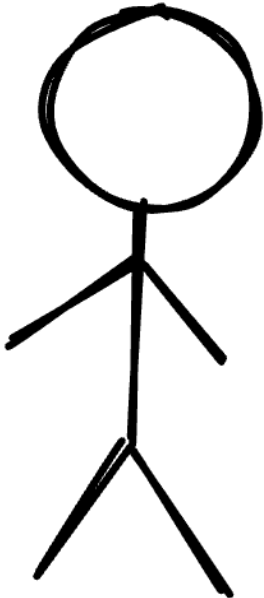
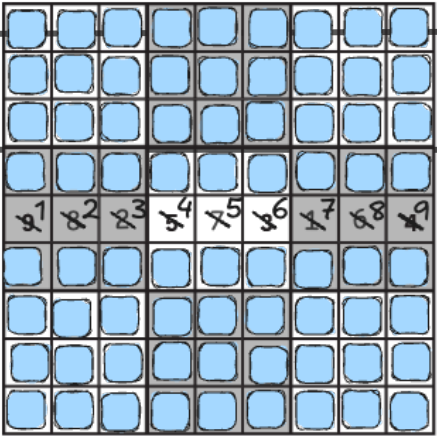
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| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
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| 4 | 5 | 9 | 3 | 1 | 7 | 2 | 8 | 6 |



Compromiso al  
tablero permutado

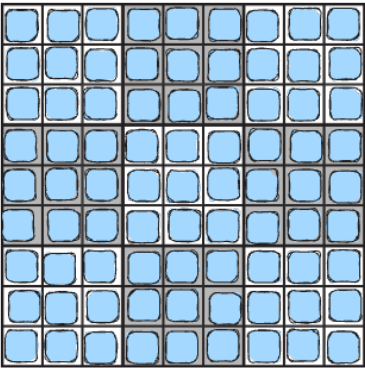


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Fila  
Columna  
Bloque  
Permutación

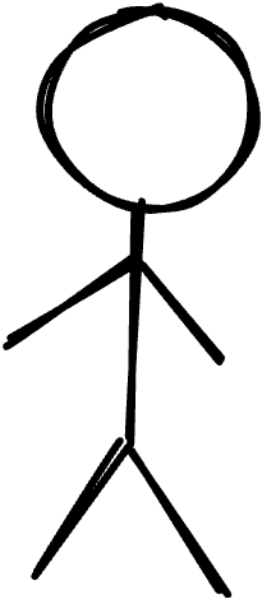
# Ejemplo: Sudoku 3/3



Compromiso al  
tablero permutado

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 | 4 | 6 | 3 | 8 | 9 | 7 | 1 |
| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
| 8 | 7 | 1 | 2 | 4 | 6 | 5 | 9 | 3 |
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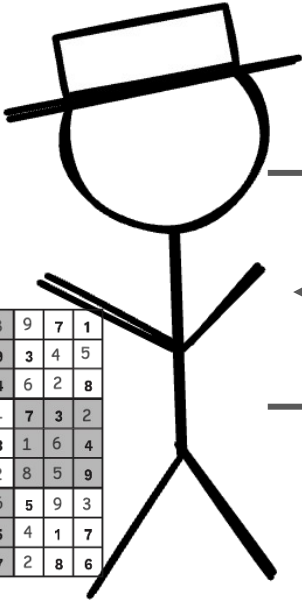
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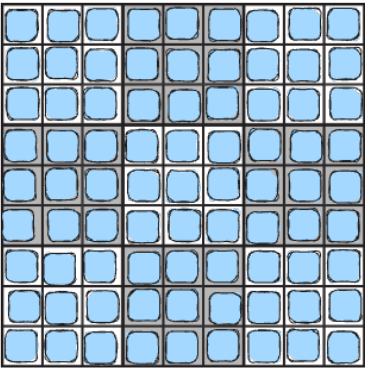
Fila  
Columna  
Bloque  
Permutación

# Ejemplo: Sudoku 3/3

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 2 | 4 | 6 | 3 | 8 | 9 | 7 | 1 |
| 7 | 6 | 8 | 1 | 2 | 9 | 3 | 4 | 5 |
| 1 | 9 | 3 | 7 | 5 | 4 | 6 | 2 | 8 |
| 6 | 4 | 5 | 9 | 8 | 1 | 7 | 3 | 2 |
| 9 | 8 | 2 | 5 | 7 | 3 | 1 | 6 | 4 |
| 3 | 1 | 7 | 4 | 6 | 2 | 8 | 5 | 9 |
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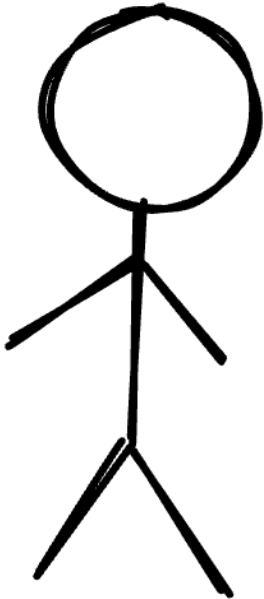


Compromiso al  
tablero permutado



$\alpha$

|             |             |             |          |  |          |          |          |             |
|-------------|-------------|-------------|----------|--|----------|----------|----------|-------------|
| $\phi_4$    | $\phi_6$    |             | $\phi_8$ |  |          |          | $\phi_5$ | $\phi_7$    |
| $\phi_7$    | $\phi_{10}$ |             | $\phi_7$ |  | $\phi_1$ | $\phi_6$ |          |             |
|             |             | $\phi_6$    |          |  | $\phi_9$ |          |          | $\phi_3$    |
| $\phi_{10}$ |             |             |          |  |          | $\phi_5$ | $\phi_6$ |             |
| $\phi_1$    |             |             | $\phi_4$ |  | $\phi_6$ |          |          | $\phi_9$    |
|             | $\phi_7$    | $\phi_5$    |          |  |          |          |          | $\phi_1$    |
| $\phi_3$    |             |             | $\phi_3$ |  |          | $\phi_4$ |          |             |
|             |             | $\phi_{10}$ | $\phi_3$ |  | $\phi_4$ |          | $\phi_7$ | $\phi_5$    |
| $\phi_9$    | $\phi_4$    | $\phi_{10}$ |          |  | $\phi_5$ |          | $\phi_3$ | $\phi_{10}$ |



Fila  
Columna  
Bloque  
Permutación

Tornado Cash

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## Overview

ETH BALANCE

0.784117837097022634 ETH

ETH VALUE

\$3,247.69 (@ \$4,141.83/ETH)

TOKEN HOLDINGS

>\$244,255.59 (>401 Tokens)

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PRIVATE NAME TAGS

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TRANSACTIONS SENT

Latest: 1 day ago ↗ First: 10 yrs 13 days ago ↗

FUNDED BY

[Vb 2](#) | 10 yrs 16 days ago

## Multichain Info

\$3,763,088.46 (Multichain Portfolio)

18 addresses found via [Blockscan](#) ▾

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[Internal Transactions](#)

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⌵ Latest 25 from a total of 73,373 transactions (+2 Pending)

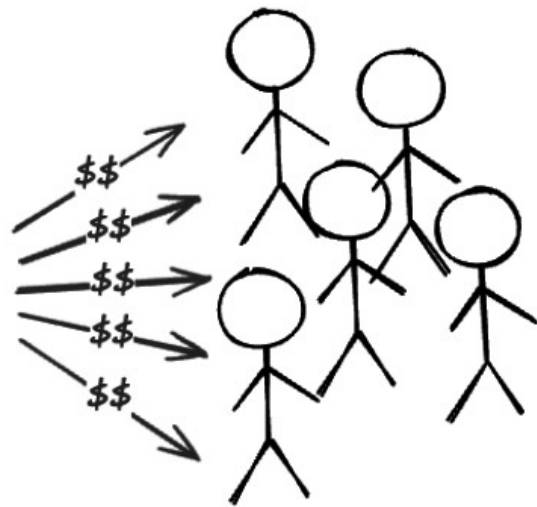
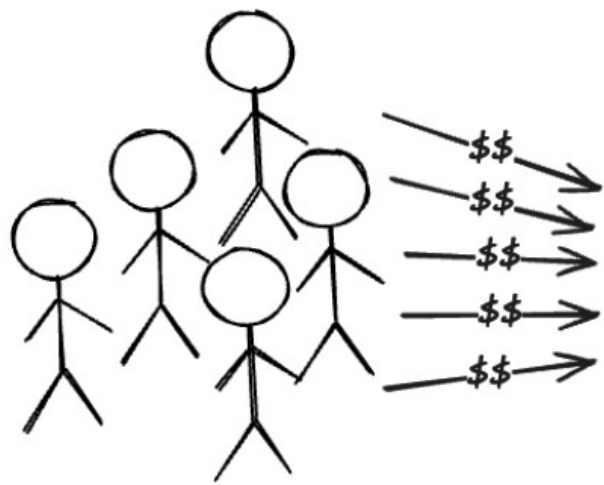
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| Transaction Hash                 | Method                   | Block     | Age         | From                                   | To                                                        | Amount     | Txn Fee   |
|----------------------------------|--------------------------|-----------|-------------|----------------------------------------|-----------------------------------------------------------|------------|-----------|
| <a href="#">0xd437fbed519...</a> | <a href="#">Transfer</a> | (pending) | 9 days ago  | <a href="#">0x93072e59...b2a09218e</a> | <a href="#">IN</a> <a href="#">0xd8dA6BF2...37aA96045</a> | 0.0001 ETH | (Pending) |
| <a href="#">0x2191cfcb192...</a> | <a href="#">Transfer</a> | (pending) | 23 days ago | <a href="#">0x17BfAE72...92b956F4B</a> | <a href="#">IN</a> <a href="#">0xd8dA6BF2...37aA96045</a> | 0.0005 ETH | (Pending) |

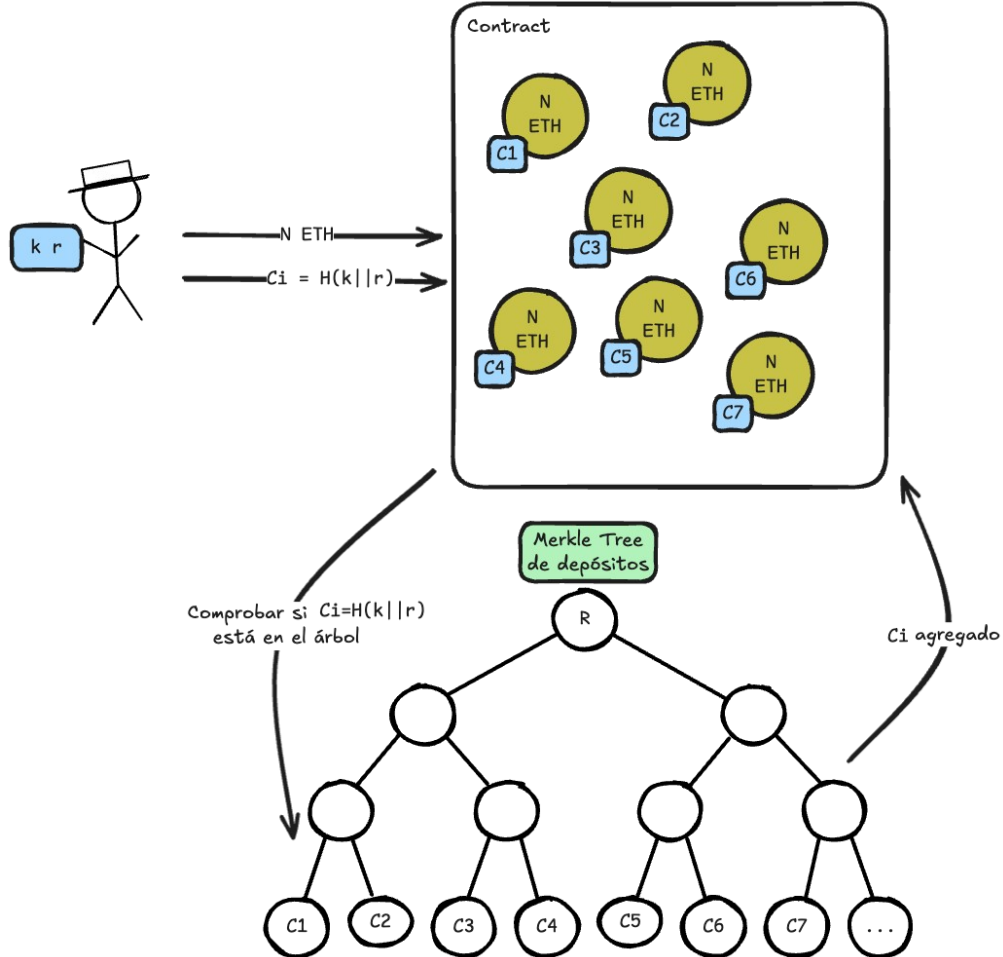


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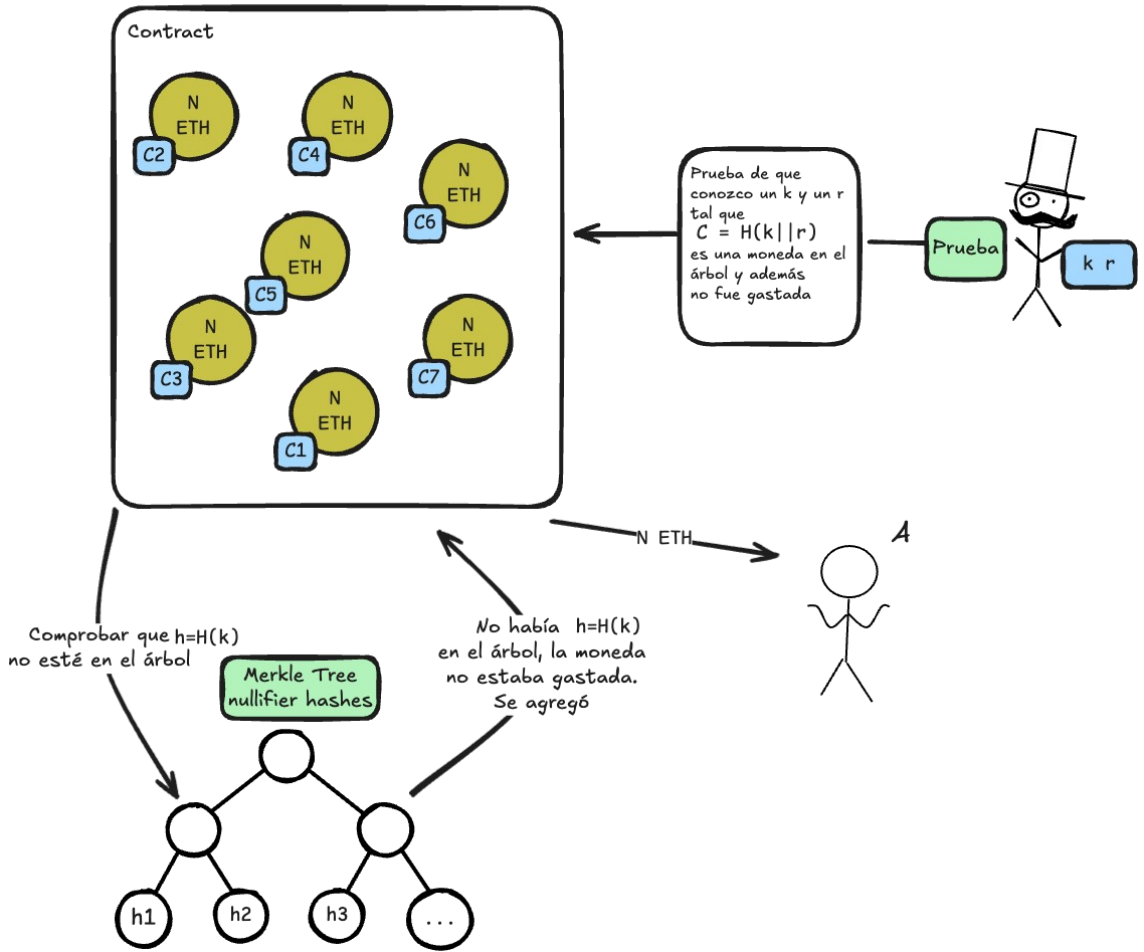
|  Home Blockchain Tokens NFTs Resources Developers More Sign In |        |            |              |                             |      |                                        |                |               |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|--------------|-----------------------------|------|----------------------------------------|----------------|---------------|--|
| Transaction Hash                                                                                                                                | Type ▾ | Method ⓘ ▾ | Age ▾        | From ▾                      | OR ▾ | To ▾                                   | Amount ▾       | Asset ▾       |  |
| <a href="#">0xb3e9f5f2418...</a>                                                                                                                | ERC-20 | Transfer   | 37 days ago  | <a href="#">vitalik.eth</a> | →    | <a href="#">0x406d0814...208C2Cdbc</a> | 4,242          | USDC (USDC) ▾ |  |
| <a href="#">0x2dc127e4fc4...</a>                                                                                                                | ERC-20 | Transfer   | 132 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">0x1810c87a...c03EFF10E</a> | 100,000        | USDC (USDC) ▾ |  |
| <a href="#">0x9f5f946970f...</a>                                                                                                                | ERC-20 | Transfer   | 179 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">0x406d0814...208C2Cdbc</a> | 4,133          | USDC (USDC) ▾ |  |
| <a href="#">0xee0fefad74c...</a>                                                                                                                | ERC-20 | Transfer   | 246 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">0x1810c87a...c03EFF10E</a> | 70,000         | USDC (USDC) ▾ |  |
| <a href="#">0xd250103501...</a>                                                                                                                 | ERC-20 | Transfer   | 281 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">Kanro 1</a>                | 21,211.050963  | USDC (USDC) ▾ |  |
| <a href="#">0x9814f4bcbe1...</a>                                                                                                                | ERC-20 | Transfer   | 281 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">Kanro 1</a>                | 518,387.316693 | USDC (USDC) ▾ |  |
| <a href="#">0x49ff9a7e540...</a>                                                                                                                | ERC-20 | Transfer   | 281 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">Kanro 1</a>                | 98,749.57226   | USDC (USDC) ▾ |  |
| <a href="#">0x6f4d4503799...</a>                                                                                                                | ERC-20 | Transfer   | 429 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">0x3AF520a3...C372d9F16</a> | 3,806.071279   | USDC (USDC) ▾ |  |
| <a href="#">0x9a4a848846...</a>                                                                                                                 | ERC-20 | Transfer   | 437 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">0x2Ba55F98...D305915B8</a> | 3,085.54033    | USDC (USDC) ▾ |  |
| <a href="#">0xd656d1125a...</a>                                                                                                                 | ERC-20 | Transfer   | 463 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">0x496C44b3...4095312DD</a> | 100            | USDC (USDC) ▾ |  |
| <a href="#">0xaf43d7d4204...</a>                                                                                                                | ERC-20 | Transfer   | 466 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">0x2Bc9d7b1...1d8aD9b1D</a> | 75             | USDC (USDC) ▾ |  |
| <a href="#">0xa080e94bbe...</a>                                                                                                                 | ERC-20 | Transfer   | 517 days ago | <a href="#">vitalik.eth</a> | →    | <a href="#">labs.taiko.eth</a>         | 2,000          | USDC (USDC) ▾ |  |



## Depósito



# Retiro



(zk)SNARKs

# (zk)SNARKs

(zero-knowledge)

# (zk)SNARKs

(zero-knowledge)

Succinct

# (zk)SNARKs

(zero-knowledge)

Succinct

Non-interactive



# (zk)SNARKs

(zero-knowledge)

Succinct

Non-interactive

ARguments of

# (zk)SNARKs

(zero-knowledge)

Succinct

Non-interactive

ARguments of

Knowledge

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Non-interactive

ARguments of

Knowledge

Permiten probar que conocemos una solución a un sistema de ecuaciones.

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ARguments of

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**Aritmetización**

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programas

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programas —→ **Aritmetización**

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programas  $\longrightarrow$  **Aritmetización**  $\longrightarrow$

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programas  $\longrightarrow$  **Aritmetización**  $\longrightarrow$  sistemas de ecuaciones



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programas  $\longrightarrow$  **Aritmetización**  $\longrightarrow$  sistemas de ecuaciones

PLONK: sistema de ecuaciones cuadráticas sobre  $F_p$

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programas  $\longrightarrow$  **Aritmetización**  $\longrightarrow$  sistemas de ecuaciones

PLONK: sistema de ecuaciones cuadráticas sobre  $F_p$

Todo problema NP puede escribirse como un sistema de ecuaciones cuadráticas

# (zk)SNARKs

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ARguments of

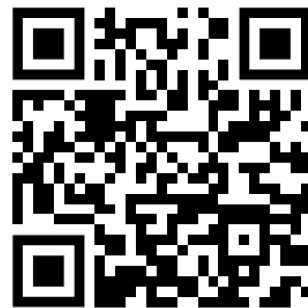
Knowledge

Permiten probar que conocemos una solución a un sistema de ecuaciones.

programas  $\longrightarrow$  **Aritmetización**  $\longrightarrow$  sistemas de ecuaciones

PLONK: sistema de ecuaciones cuadráticas sobre  $F_p$

Todo problema NP puede escribirse como un sistema de ecuaciones cuadráticas



Introduction to Programmable  
Cryptography - 0xPARC

```

28 // Verifies that merkle proof is correct for given merkle root and a leaf
29 // pathIndices input is an array of 0/1 selectors telling whether given pathElement is on the left or right side of merkle path
30 template MerkleTreeChecker(levels) {
31     signal input leaf;
32     signal input root;
33     signal input pathElements[levels];
34     signal input pathIndices[levels];
35
36     component selectors[levels];
37     component hashers[levels];
38
39     for (var i = 0; i < levels; i++) {
40         selectors[i] = DualMux();
41         selectors[i].in[0] <== i == 0 ? leaf : hashers[i - 1].hash;
42         selectors[i].in[1] <== pathElements[i];
43         selectors[i].s <== pathIndices[i];
44
45         hashers[i] = HashLeftRight();
46         hashers[i].left <== selectors[i].out[0];
47         hashers[i].right <== selectors[i].out[1];
48     }
49
50     root == hashers[levels - 1].hash;
51 }

```

[github.com/tornadocash/tornado-core/blob/master/circuits/merkleTree.circom](https://github.com/tornadocash/tornado-core/blob/master/circuits/merkleTree.circom)

Gracias!